Arboring Widths - Gymnosperms

Tree ID Time Series for Gymnosperms

```
#reshape data from wider to longer format

library(tidyverse)
```

```
Warning: package 'tidyverse' was built under R version 4.5.2

Warning: package 'ggplot2' was built under R version 4.5.2

Warning: package 'tibble' was built under R version 4.5.2

Warning: package 'tidyr' was built under R version 4.5.2

Warning: package 'readr' was built under R version 4.5.2

Warning: package 'purrr' was built under R version 4.5.2

Warning: package 'dplyr' was built under R version 4.5.2

Warning: package 'forcats' was built under R version 4.5.2

Warning: package 'lubridate' was built under R version 4.5.2

── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
✓ dplyr     1.1.4     ✓ readr     2.1.6
✓ forcats   1.0.1     ✓ stringr   1.5.2
✓ ggplot2   4.0.1     ✓ tibble    3.3.0
✓ lubridate 1.9.5     ✓ tidyr     1.3.2
✓ purrr     1.2.1
── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
✗ dplyr::filter() masks stats::filter()
✗ dplyr::lag()    masks stats::lag()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
errors
```

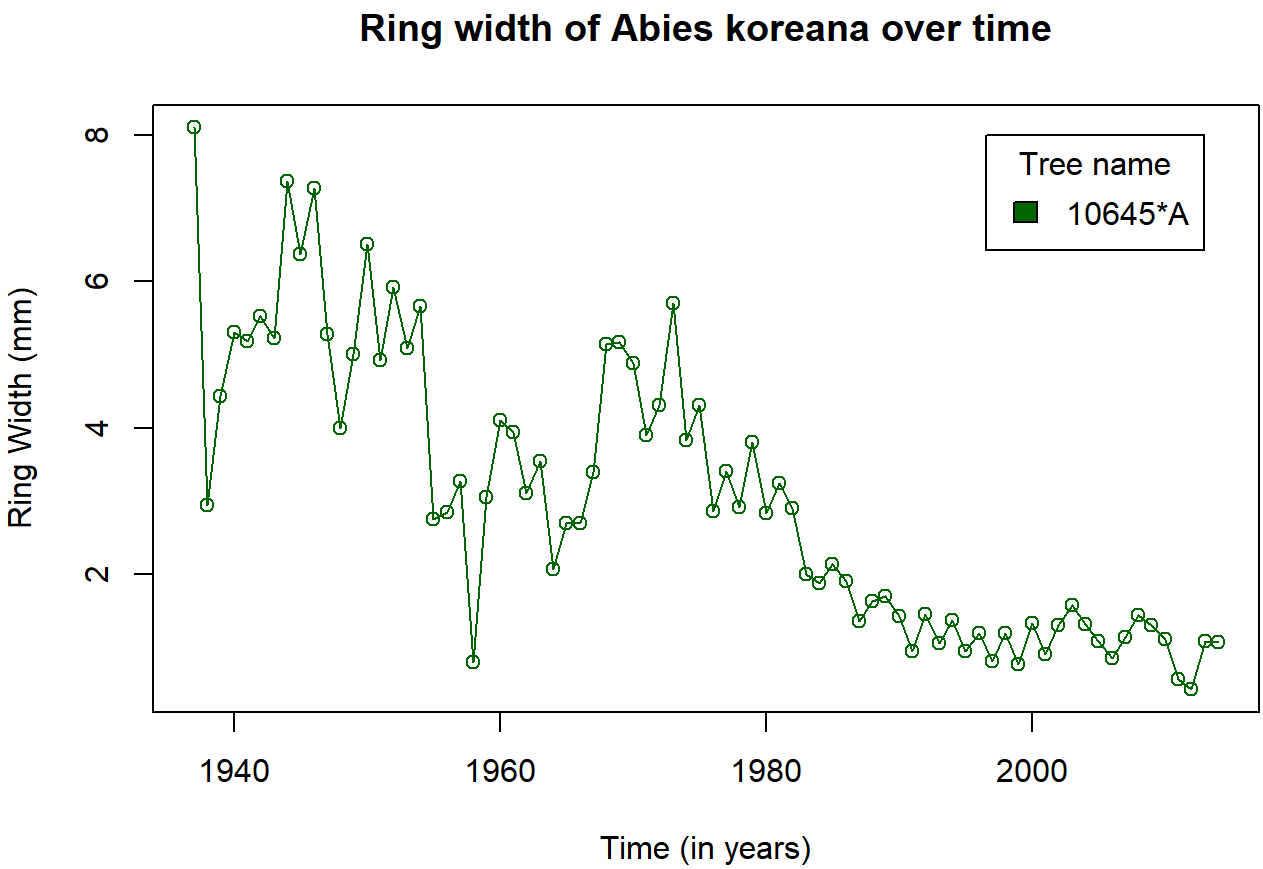
```
# Load data
gymno <- read.csv("Arboringwidthsgymnosperms.csv")


# Remove X in front of year columns by subbing "X" with nothing ""
colnames(gymno) <- sub("X", "", colnames(gymno))

# Pivot data
gymno_data_long <- pivot_longer(data = gymno, cols = "2014":"1861", names_to = "Year", values_to =
```

```r
# View data
print(gymno_data_long)
```

```
# A tibble: 13,103 × 9
   TreeName Species      YearLastRing Site  RingCount ImageName AnalysisDateTime
   <chr>    <chr>               <int> <chr>     <int> <chr>     <chr>
 1 10645*A  Abies_korea…         2014 Arno…        78 Abies_sp… 7/20/16 9:39
 2 10645*A  Abies_korea…         2014 Arno…        78 Abies_sp… 7/20/16 9:39
 3 10645*A  Abies_korea…         2014 Arno…        78 Abies_sp… 7/20/16 9:39
 4 10645*A  Abies_korea…         2014 Arno…        78 Abies_sp… 7/20/16 9:39
 5 10645*A  Abies_korea…         2014 Arno…        78 Abies_sp… 7/20/16 9:39
 6 10645*A  Abies_korea…         2014 Arno…        78 Abies_sp… 7/20/16 9:39
 7 10645*A  Abies_korea…         2014 Arno…        78 Abies_sp… 7/20/16 9:39
 8 10645*A  Abies_korea…         2014 Arno…        78 Abies_sp… 7/20/16 9:39
 9 10645*A  Abies_korea…         2014 Arno…        78 Abies_sp… 7/20/16 9:39
10 10645*A  Abies_korea…         2014 Arno…        78 Abies_sp… 7/20/16 9:39
# ℹ 13,093 more rows
# ℹ 2 more variables: Year <chr>, Ringwidth <dbl>
```

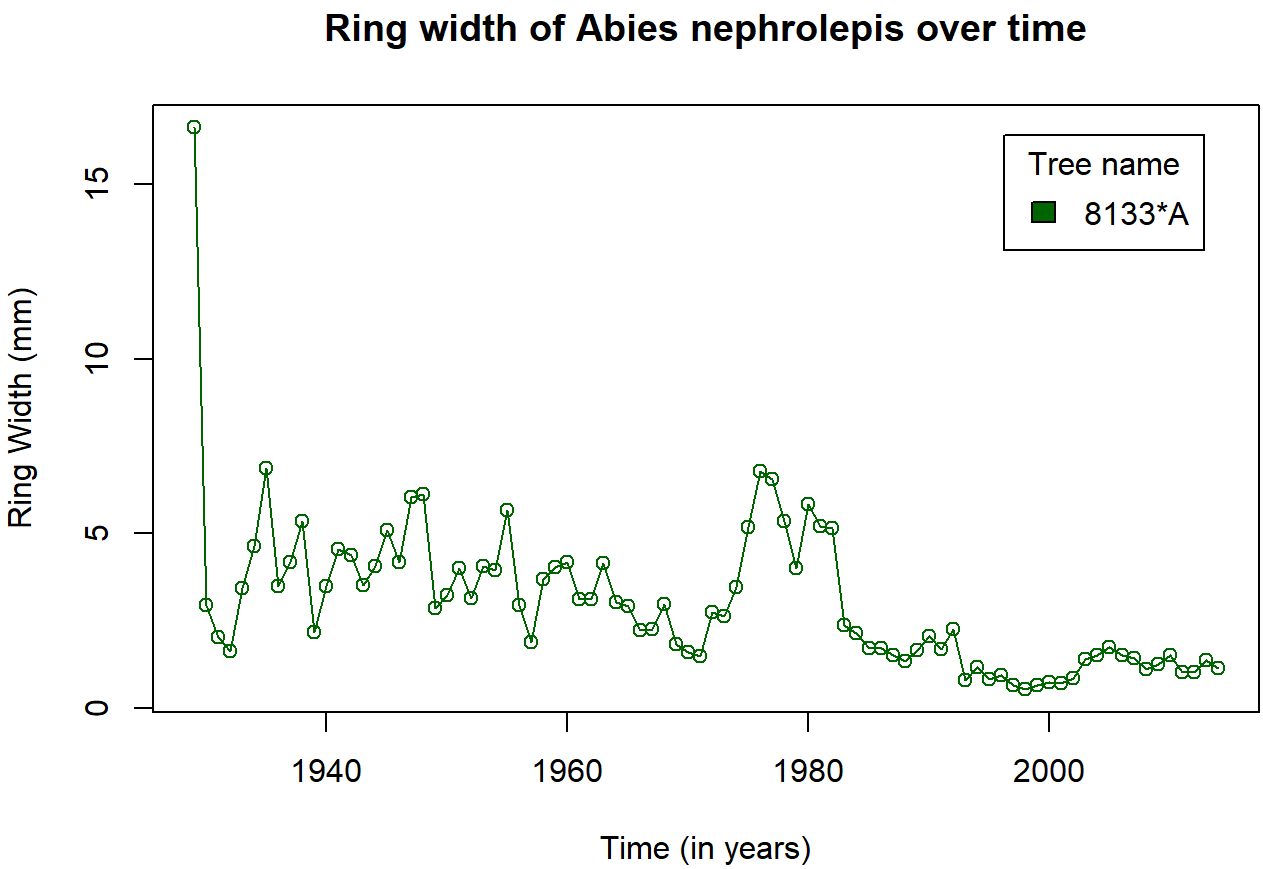
Abies koreana

```r
koreana <- subset(gymno_data_long, TreeName == "10645*A")

plot(koreana$Year, y = koreana$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", y

legend("topright", inset = .05, title = "Tree name", c("10645*A"), fill = c("darkgreen"), horiz =
```



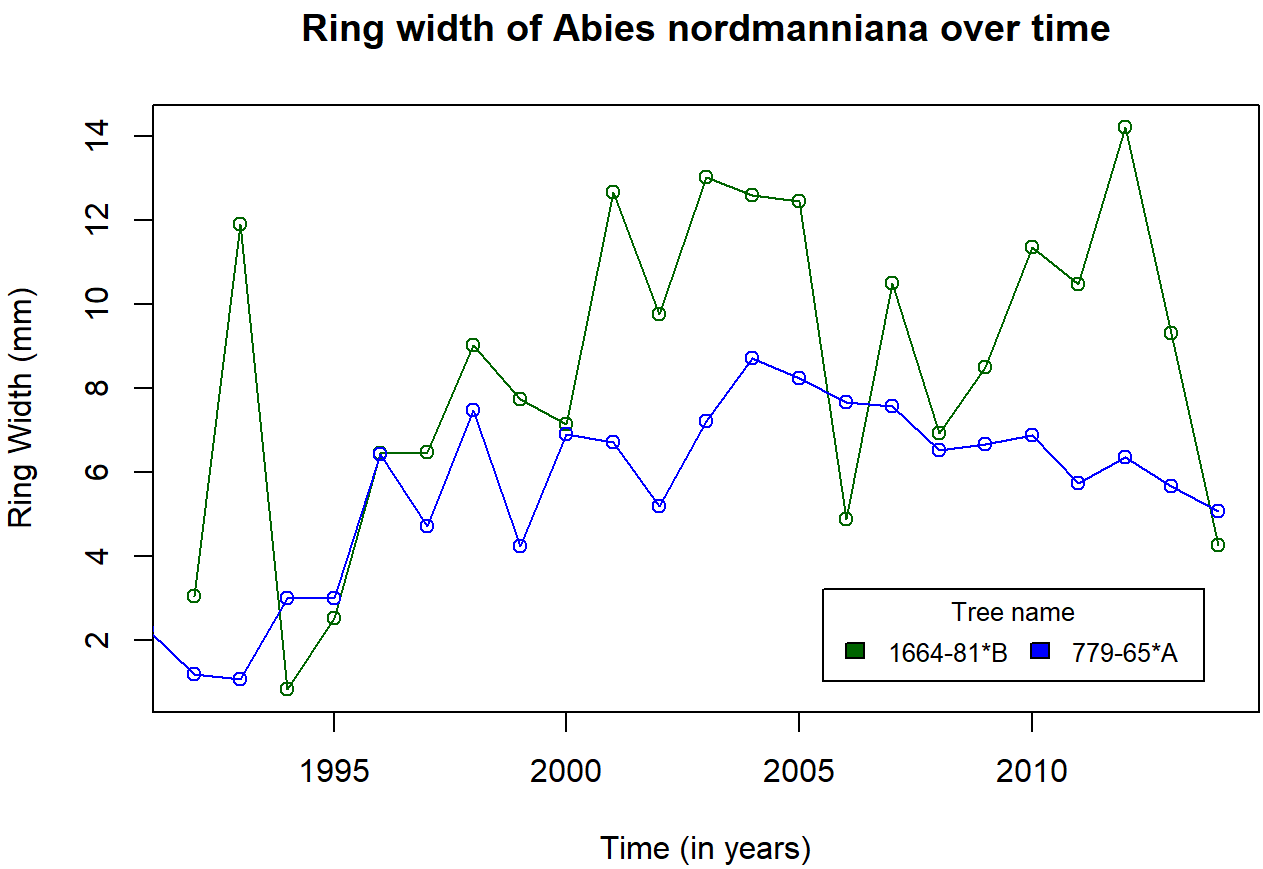
Abies nephrolepsis

```
nephrolepsis <- subset(gymno_data_long, TreeName == "8133*A")

plot(nephrolepsis$Year, y = nephrolepsis$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in

legend("topright", inset = .05, title = "Tree name", c("8133*A"), fill = c("darkgreen"), horiz =
```



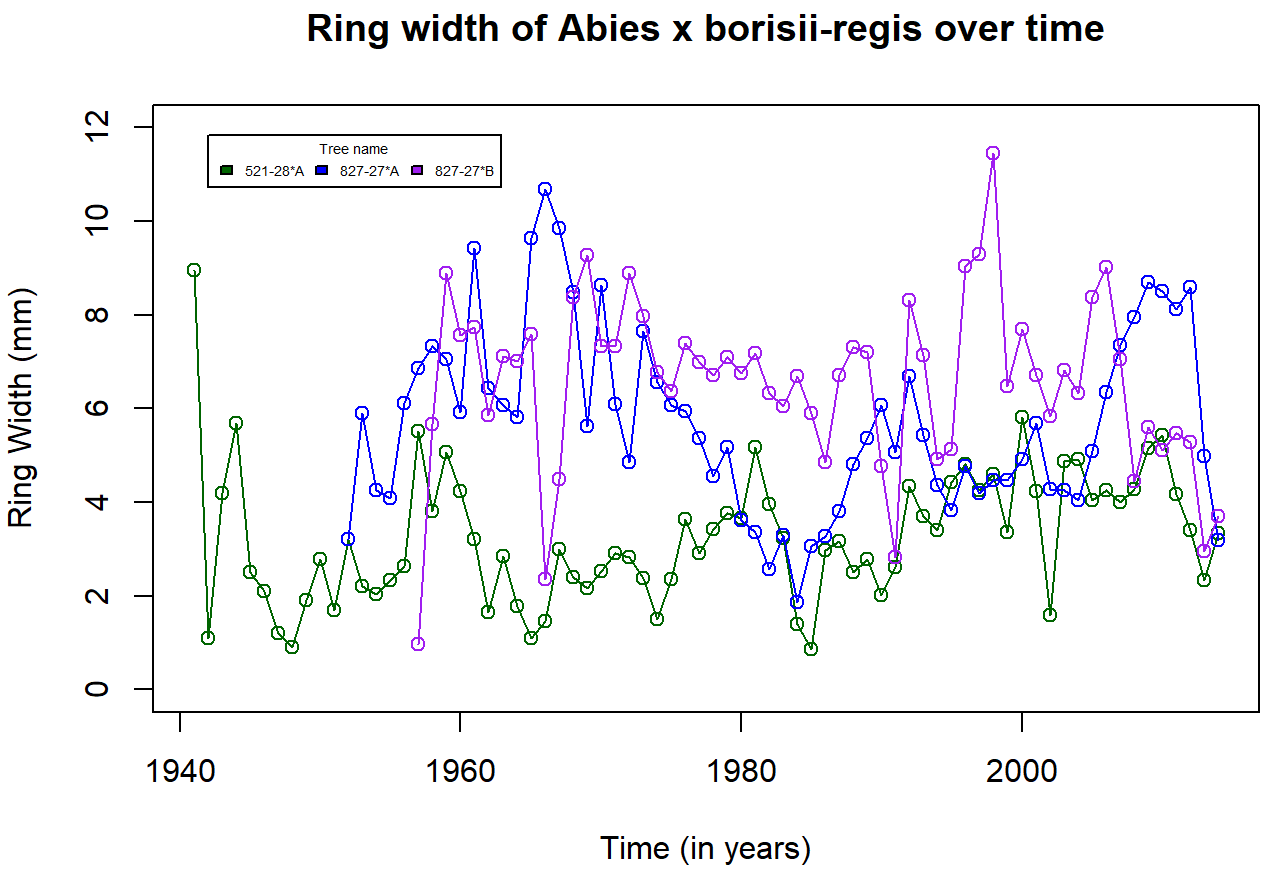
Abies nordmanniana

```
nord1 <- subset(gymno_data_long, TreeName == "1664-81*B")

nord2 <- subset(gymno_data_long, TreeName == "779-65*A")


plot(nord1$Year, y = nord1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", ylab =
lines(nord2$Year, y = nord2$Ringwidth, type = "o", col = "blue")

legend("bottomright", inset = .05, title = "Tree name", c("1664-81*B", "779-65*A"), fill = c("darl
```



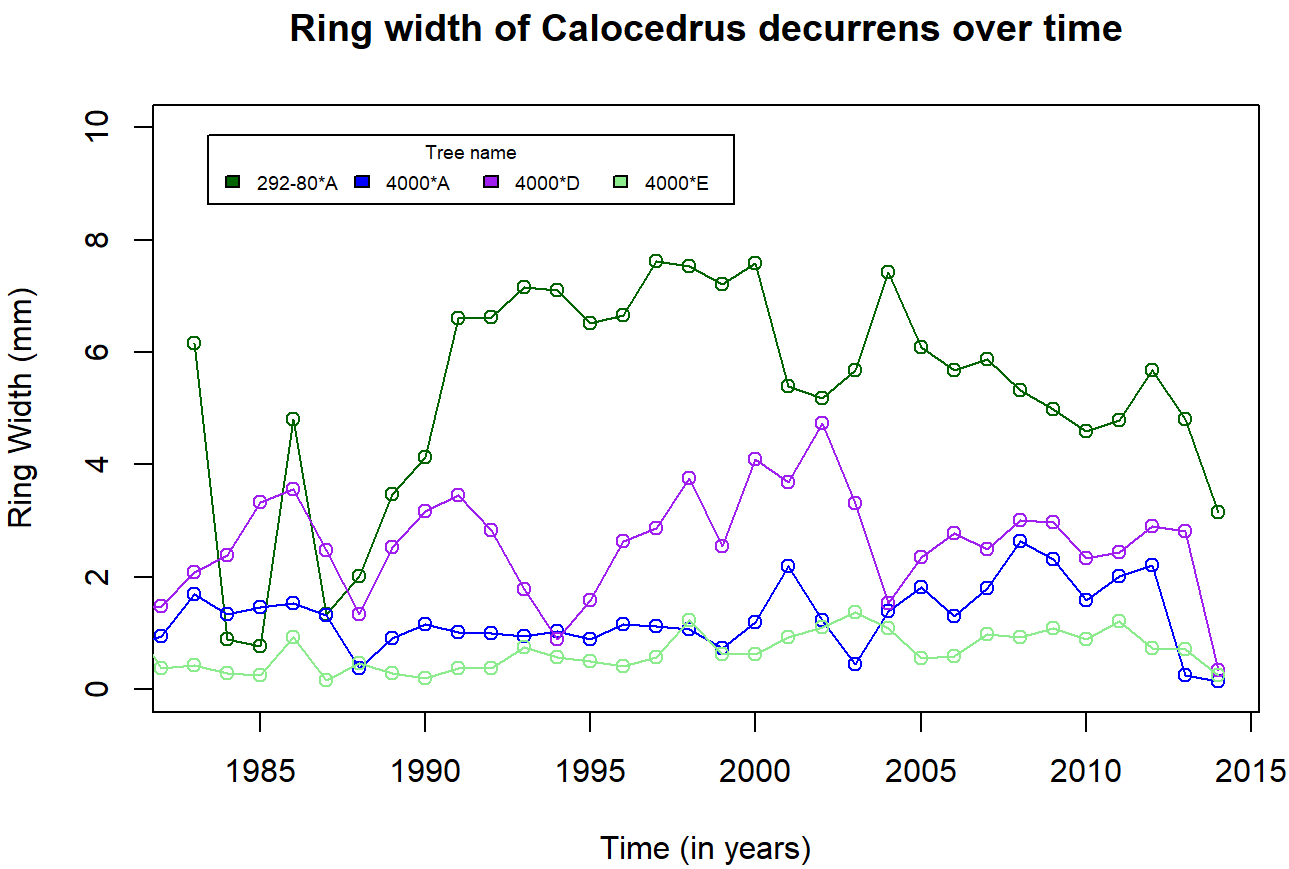
Abies x borisii-regis

```
borisii1 <- subset(gymno_data_long, TreeName == "521-28*A")

borisii2 <- subset(gymno_data_long, TreeName == "827-27*A")

borisii3 <- subset(gymno_data_long, TreeName == "827-27*B")


plot(borisii1$Year, y = borisii1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)",
lines(borisii2$Year, y = borisii2$Ringwidth, type = "o", col = "blue")
lines(borisii3$Year, y = borisii3$Ringwidth, type = "o", col = "purple")

legend("topleft", inset = .05, title = "Tree name", c("521-28*A", "827-27*A", "827-27*B"), fill =
```



Calocedrus decurrens

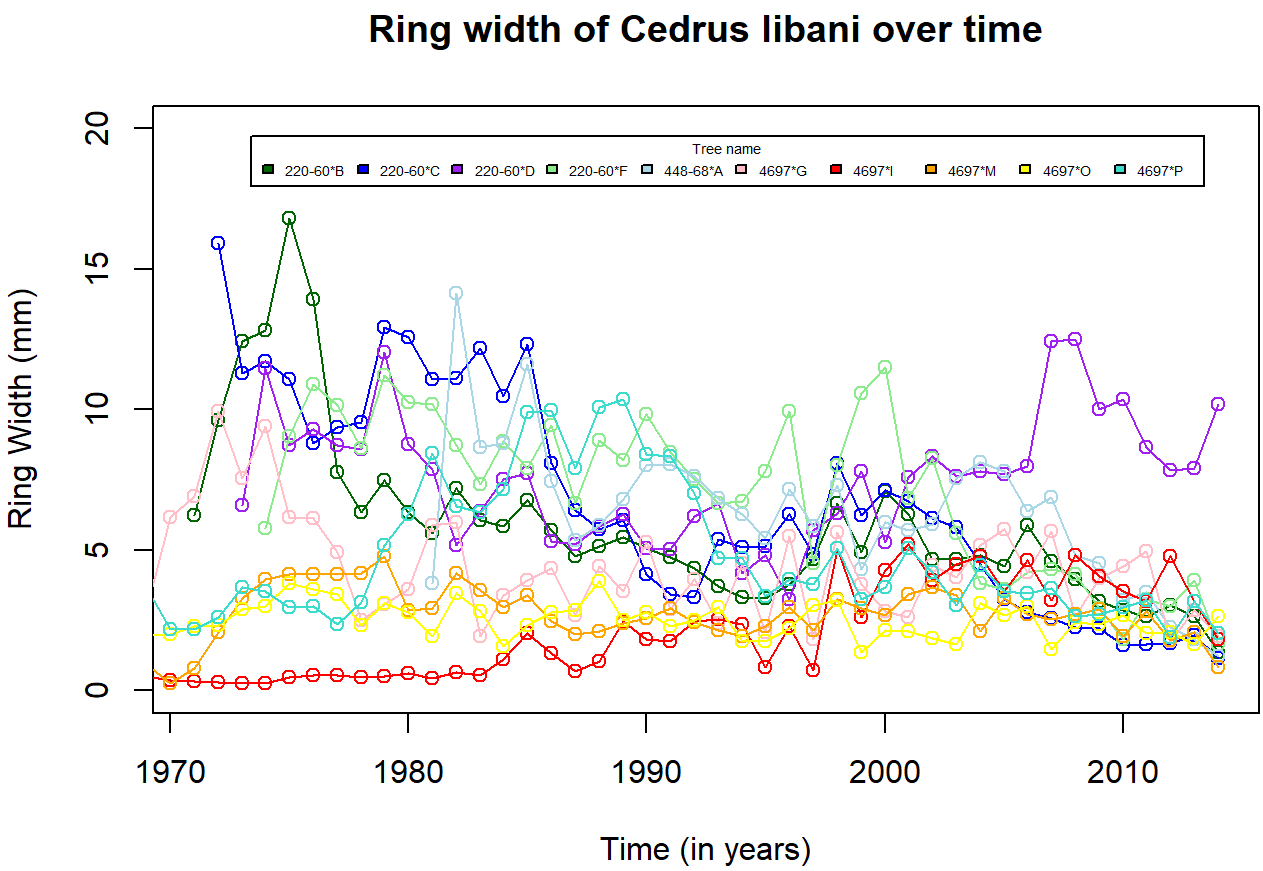
```
decurrens1 <- subset(gymno_data_long, TreeName == "292-80*A")

decurrens2 <- subset(gymno_data_long, TreeName == "4000*A")

decurrens3 <- subset(gymno_data_long, TreeName == "4000*D")

decurrens4 <- subset(gymno_data_long, TreeName == "4000*E")


plot(decurrens1$Year, y = decurrens1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years
lines(decurrens2$Year, y = decurrens2$Ringwidth, type = "o", col = "blue")
lines(decurrens3$Year, y = decurrens3$Ringwidth, type = "o", col = "purple")
lines(decurrens4$Year, y = decurrens4$Ringwidth, type = "o", col = "lightgreen")

legend("topleft", inset = .05, title = "Tree name", c("292-80*A", "4000*A", "4000*D", "4000*E"),
```



Cedrus libani

```
libani1 <- subset(gymno_data_long, TreeName == "220-60*B")

libani2 <- subset(gymno_data_long, TreeName == "220-60*C")

libani3 <- subset(gymno_data_long, TreeName == "220-60*D")

libani4 <- subset(gymno_data_long, TreeName == "220-60*F")

libani5 <- subset(gymno_data_long, TreeName == "448-68*A")

libani6 <- subset(gymno_data_long, TreeName == "4697*G")

libani7 <- subset(gymno_data_long, TreeName == "4697*I")

libani8 <- subset(gymno_data_long, TreeName == "4697*M")

libani9 <- subset(gymno_data_long, TreeName == "4697*O")

libani10 <- subset(gymno_data_long, TreeName == "4697*P")
```

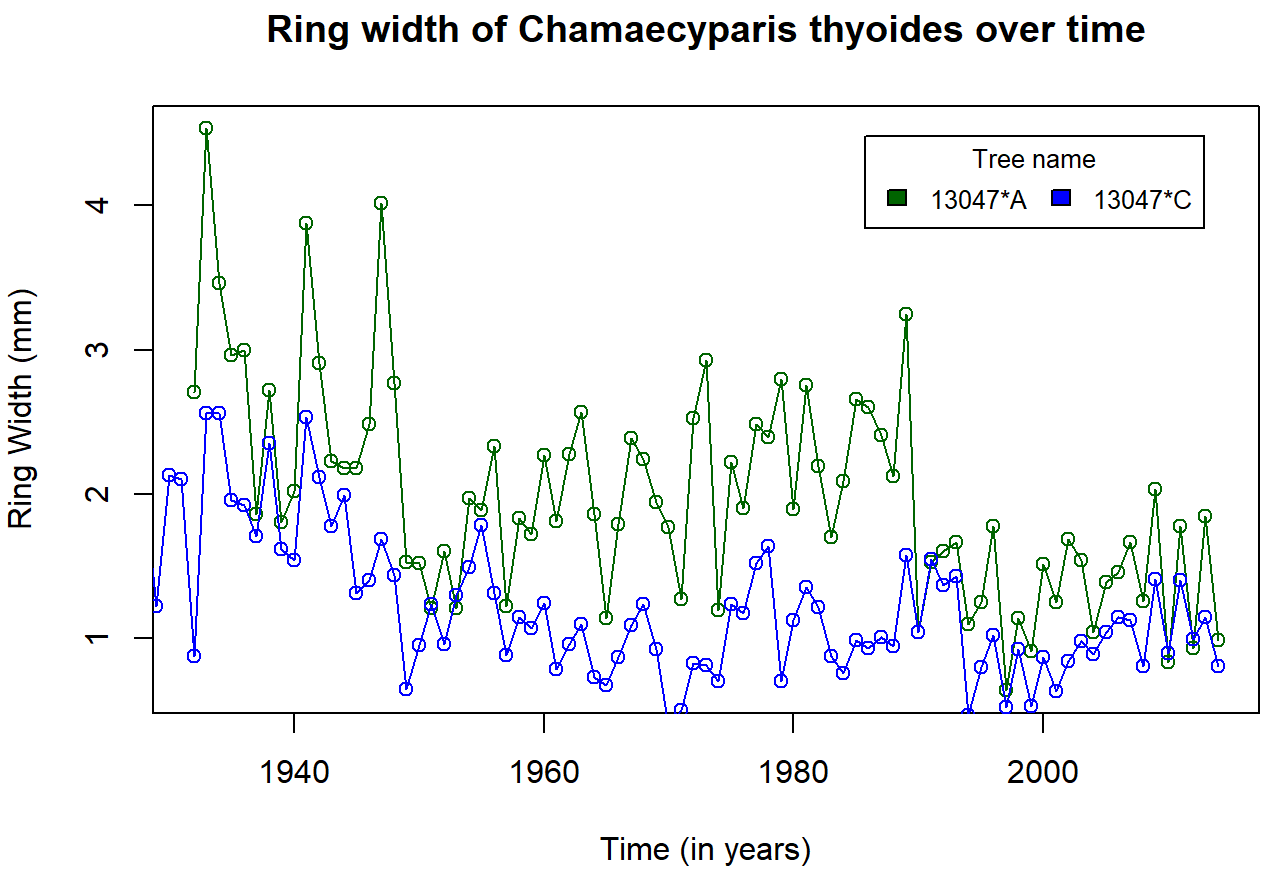
```
plot(libani1$Year, y = libani1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", y
lines(libani2$Year, y = libani2$Ringwidth, type = "o", col = "blue")
lines(libani3$Year, y = libani3$Ringwidth, type = "o", col = "purple")
lines(libani4$Year, y = libani4$Ringwidth, type = "o", col = "lightgreen")
lines(libani5$Year, y = libani5$Ringwidth, type = "o", col = "lightblue")
lines(libani6$Year, y = libani6$Ringwidth, type = "o", col = "pink")
lines(libani7$Year, y = libani7$Ringwidth, type = "o", col = "red")
lines(libani8$Year, y = libani8$Ringwidth, type = "o", col = "orange")
lines(libani9$Year, y = libani9$Ringwidth, type = "o", col = "yellow")
lines(libani10$Year, y = libani10$Ringwidth, type = "o", col = "turquoise")

legend("topright", inset = .05, title = "Tree name", c("220-60*B", "220-60*C", "220-60*D", "220-6
```



Chamaecyparis thyoides

```
chamae1 <- subset(gymno_data_long, TreeName == "13047*A")

chamae2 <- subset(gymno_data_long, TreeName == "13047*C")


plot(chamae1$Year, y = chamae1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", y
lines(chamae2$Year, y = chamae2$Ringwidth, type = "o", col = "blue")
```

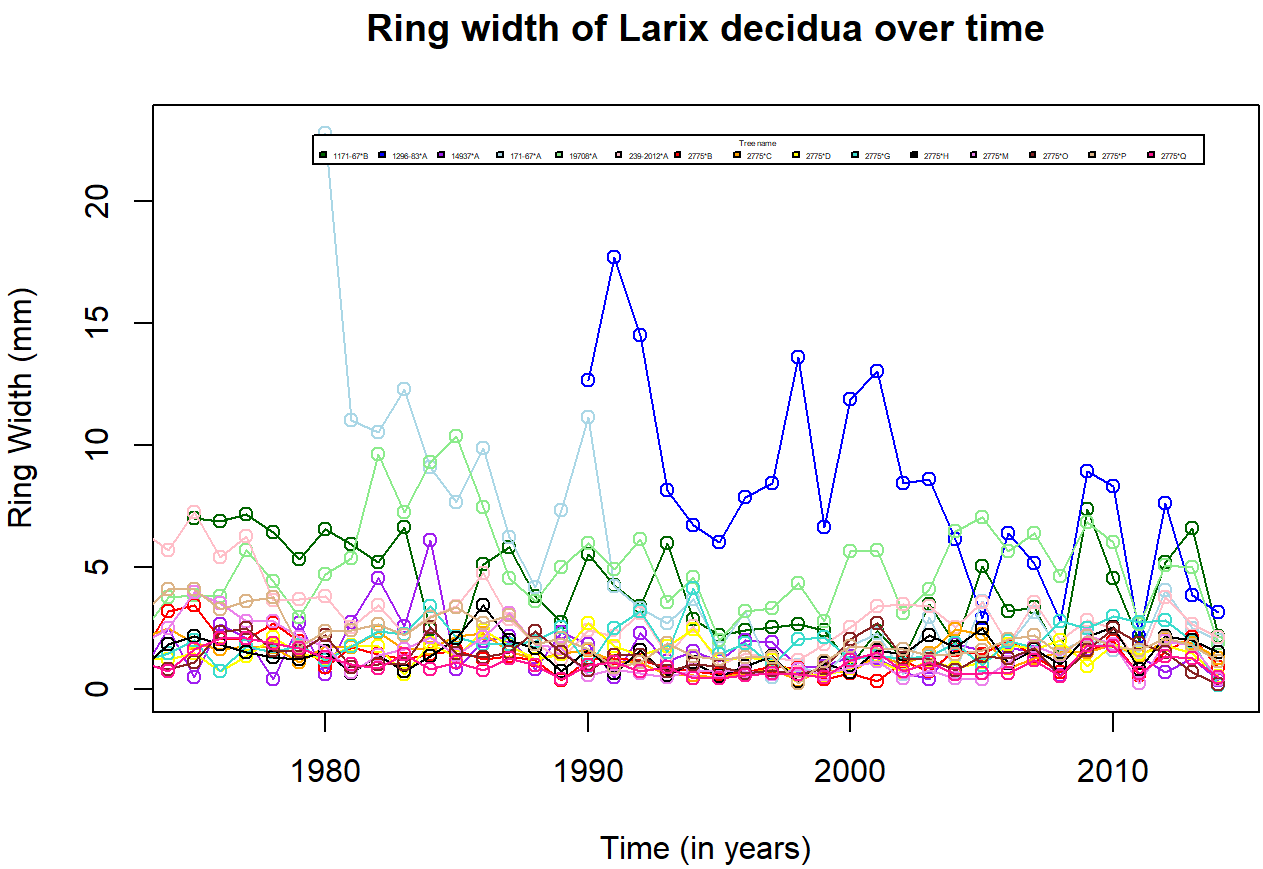
```
legend("topright", inset = .05, title = "Tree name", c("13047*A", "13047*C"), fill = c("darkgreen
```



Larix decidua

```
larix1 <- subset(gymno_data_long, TreeName == "1171-67*B")

larix2 <- subset(gymno_data_long, TreeName == "1296-83*A")

larix3 <- subset(gymno_data_long, TreeName == "14937*A")

larix4 <- subset(gymno_data_long, TreeName == "171-67*A")

larix5 <- subset(gymno_data_long, TreeName == "19708*A")

larix6 <- subset(gymno_data_long, TreeName == "239-2012*A")

larix7 <- subset(gymno_data_long, TreeName == "2775*B")

larix8 <- subset(gymno_data_long, TreeName == "2775*C")

larix9 <- subset(gymno_data_long, TreeName == "2775*D")
```

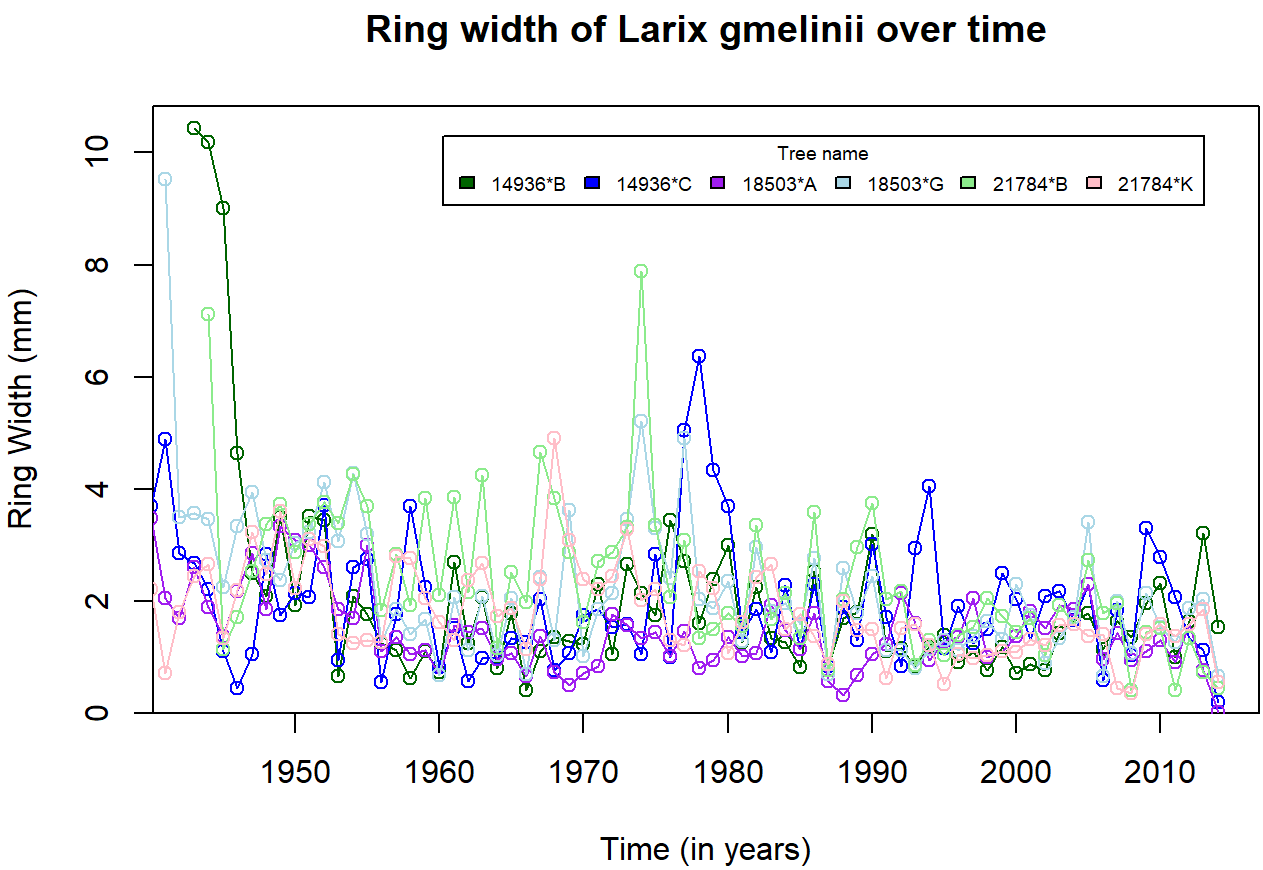
```
larix10 <- subset(gymno_data_long, TreeName == "2775*G")

larix11 <- subset(gymno_data_long, TreeName == "2775*H")

larix12 <- subset(gymno_data_long, TreeName == "2775*M")

larix13 <- subset(gymno_data_long, TreeName == "2775*O")

larix14 <- subset(gymno_data_long, TreeName == "2775*P")

larix15 <- subset(gymno_data_long, TreeName == "2775*Q")


plot(larix1$Year, y = larix1$Ringwidth, type = "o", ylim = c(0, 23), col="darkgreen", xlab= "Time
lines(larix2$Year, y = larix2$Ringwidth, type = "o", col = "blue")
lines(larix3$Year, y = larix3$Ringwidth, type = "o", col = "purple")
lines(larix4$Year, y = larix4$Ringwidth, type = "o", col = "lightblue")
lines(larix5$Year, y = larix5$Ringwidth, type = "o", col = "lightgreen")
lines(larix6$Year, y = larix6$Ringwidth, type = "o", col = "pink")
lines(larix7$Year, y = larix7$Ringwidth, type = "o", col = "red")
lines(larix8$Year, y = larix8$Ringwidth, type = "o", col = "orange")
lines(larix9$Year, y = larix9$Ringwidth, type = "o", col = "yellow")
lines(larix10$Year, y = larix10$Ringwidth, type = "o", col = "turquoise")
lines(larix11$Year, y = larix11$Ringwidth, type = "o", col = "black")
lines(larix12$Year, y = larix12$Ringwidth, type = "o", col = "violet")
lines(larix13$Year, y = larix13$Ringwidth, type = "o", col = "brown4")
lines(larix14$Year, y = larix14$Ringwidth, type = "o", col = "burlywood")
lines(larix15$Year, y = larix15$Ringwidth, type = "o", col = "deeppink")

legend("topright", inset = .05, title = "Tree name", c("1171-67*B", "1296-83*A", "14937*A", "171-
```



Larix gmelinii

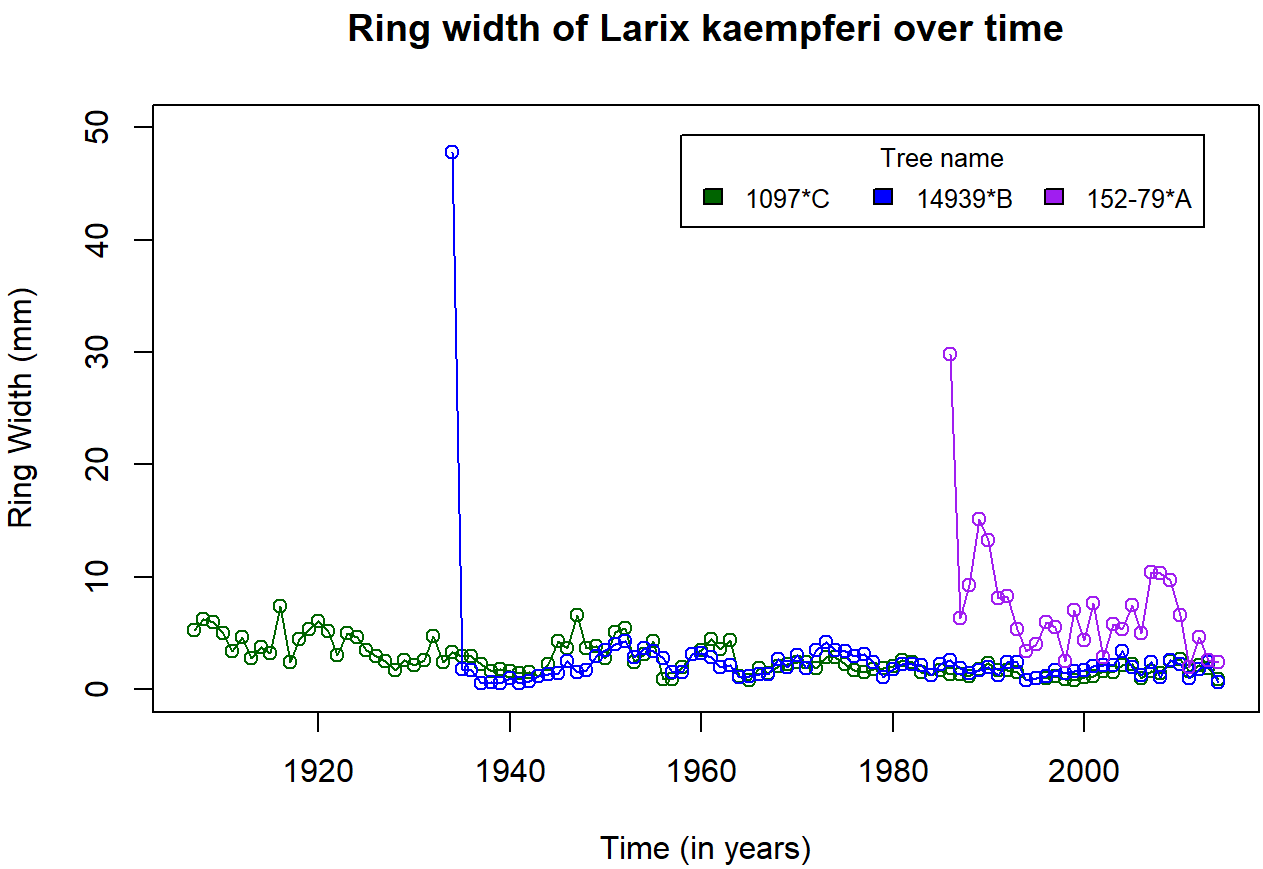
```
gmeli1 <- subset(gymno_data_long, TreeName == "14936*B")

gmeli2 <- subset(gymno_data_long, TreeName == "14936*C")

gmeli3 <- subset(gymno_data_long, TreeName == "18503*A")

gmeli4 <- subset(gymno_data_long, TreeName == "18503*G")

gmeli5 <- subset(gymno_data_long, TreeName == "21784*B")

gmeli6 <- subset(gymno_data_long, TreeName == "21784*K")


plot(gmeli1$Year, y = gmeli1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", ylal
lines(gmeli2$Year, y = gmeli2$Ringwidth, type = "o", col = "blue")
lines(gmeli3$Year, y = gmeli3$Ringwidth, type = "o", col = "purple")
lines(gmeli4$Year, y = gmeli4$Ringwidth, type = "o", col = "lightblue")
lines(gmeli5$Year, y = gmeli5$Ringwidth, type = "o", col = "lightgreen")
lines(gmeli6$Year, y = gmeli6$Ringwidth, type = "o", col = "pink")
```

```
legend("topright", inset = .05, title = "Tree name", c("14936*B", "14936*C", "18503*A", "18503*G"
```



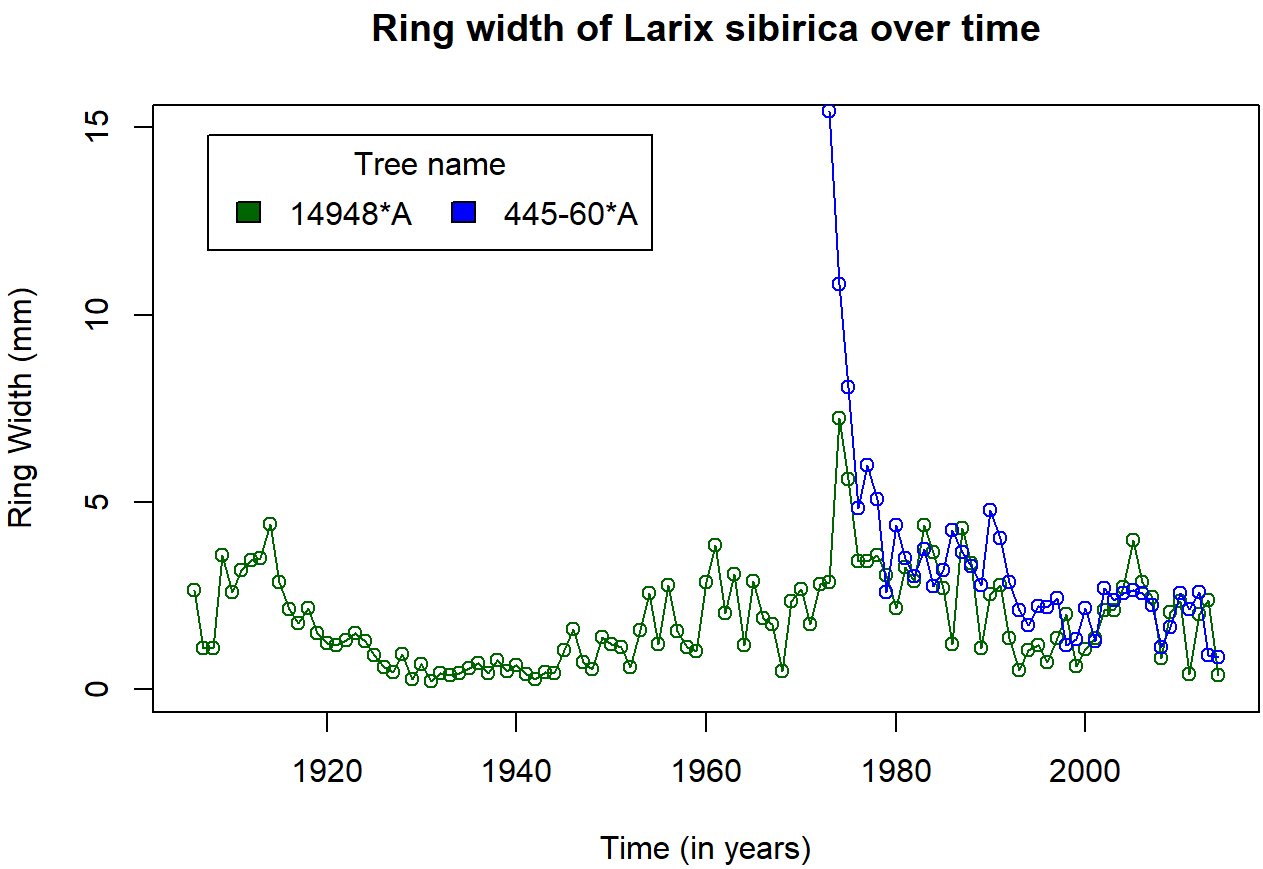
Larix kaempferi

```
kaemp1 <- subset(gymno_data_long, TreeName == "1097*C")

kaemp2 <- subset(gymno_data_long, TreeName == "14939*B")

kaemp3 <- subset(gymno_data_long, TreeName == "152-79*A")


plot(kaemp1$Year, y = kaemp1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", ylal
lines(kaemp2$Year, y = kaemp2$Ringwidth, type = "o", col = "blue")
lines(kaemp3$Year, y = kaemp3$Ringwidth, type = "o", col = "purple")

legend("topright", inset = .05, title = "Tree name", c("1097*C", "14939*B", "152-79*A"), fill = c
```



Larix sibirica

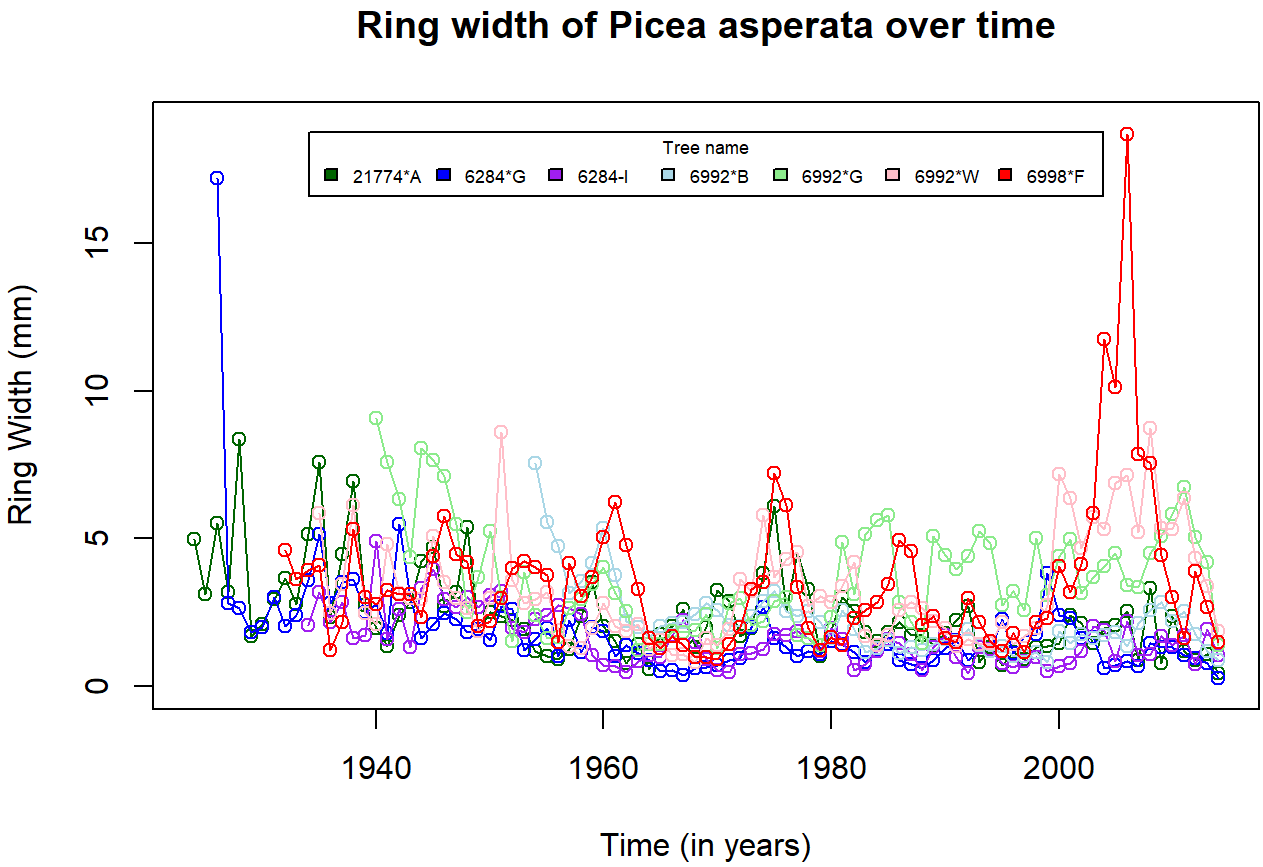
```
sibi1 <- subset(gymno_data_long, TreeName == "14948*A")

sibi2 <- subset(gymno_data_long, TreeName == "445-60*A")


plot(sibi1$Year, y = sibi1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", ylab =
lines(sibi2$Year, y = sibi2$Ringwidth, type = "o", col = "blue")

legend("topleft", inset = .05, title = "Tree name", c("14948*A", "445-60*A"), fill = c("darkgreen
```



Picea asperata

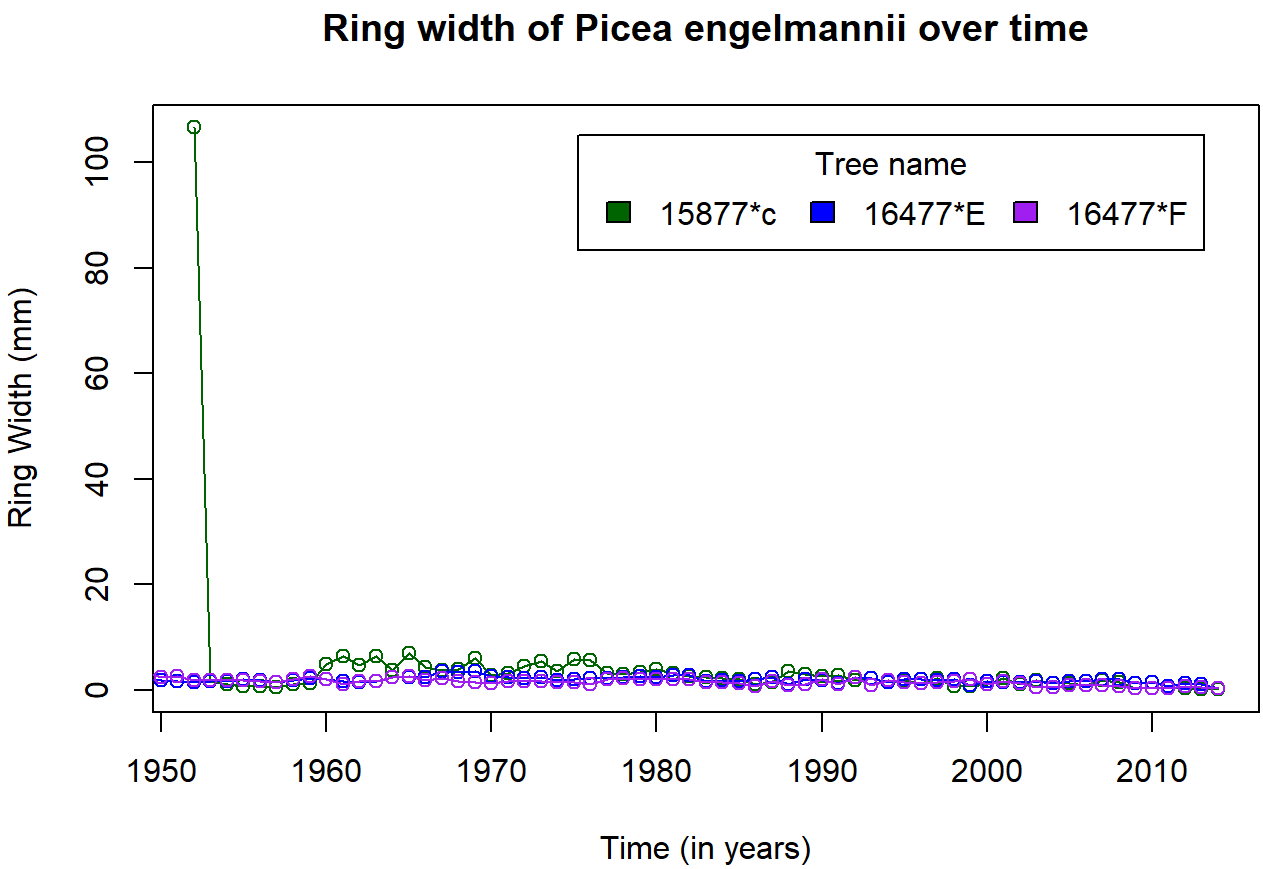
```
aspera1 <- subset(gymno_data_long, TreeName == "21774*A")

aspera2 <- subset(gymno_data_long, TreeName == "6284*G")

aspera3 <- subset(gymno_data_long, TreeName == "6284-I")

aspera4 <- subset(gymno_data_long, TreeName == "6992*B")

aspera5 <- subset(gymno_data_long, TreeName == "6992*G")

aspera6 <- subset(gymno_data_long, TreeName == "6992*W")

aspera7 <- subset(gymno_data_long, TreeName == "6998*F")


plot(aspera1$Year, y = aspera1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", y
lines(aspera2$Year, y = aspera2$Ringwidth, type = "o", col = "blue")
lines(aspera3$Year, y = aspera3$Ringwidth, type = "o", col = "purple")
lines(aspera4$Year, y = aspera4$Ringwidth, type = "o", col = "lightblue")
lines(aspera5$Year, y = aspera5$Ringwidth, type = "o", col = "lightgreen")
lines(aspera6$Year, y = aspera6$Ringwidth, type = "o", col = "pink")
```

```
lines(aspera7$Year, y = aspera7$Ringwidth, type = "o", col = "red")


legend("top", inset = .05, title = "Tree name", c("21774*A", "6284*G", "6284-I", "6992*B", "6992*(
```



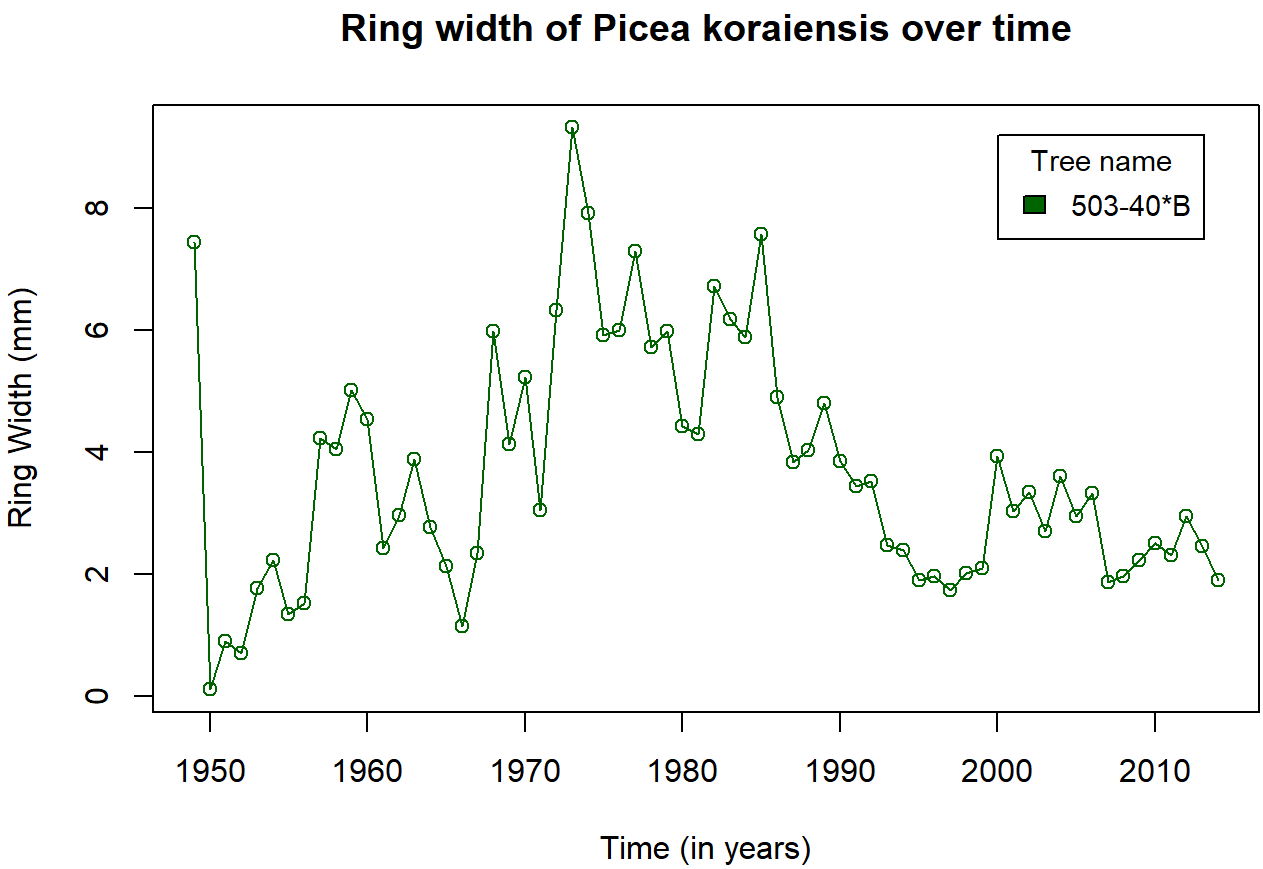
Picea engelmannii

```
engel1 <- subset(gymno_data_long, TreeName == "15877*c")

engel2 <- subset(gymno_data_long, TreeName == "16477*E")

engel3 <- subset(gymno_data_long, TreeName == "16477*F")


plot(engel1$Year, y = engel1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", ylal
lines(engel2$Year, y = engel2$Ringwidth, type = "o", col = "blue")
lines(engel3$Year, y = engel3$Ringwidth, type = "o", col = "purple")

legend("topright", inset = .05, title = "Tree name", c("15877*c", "16477*E", "16477*F"), fill = c
```



Picea koraiensis

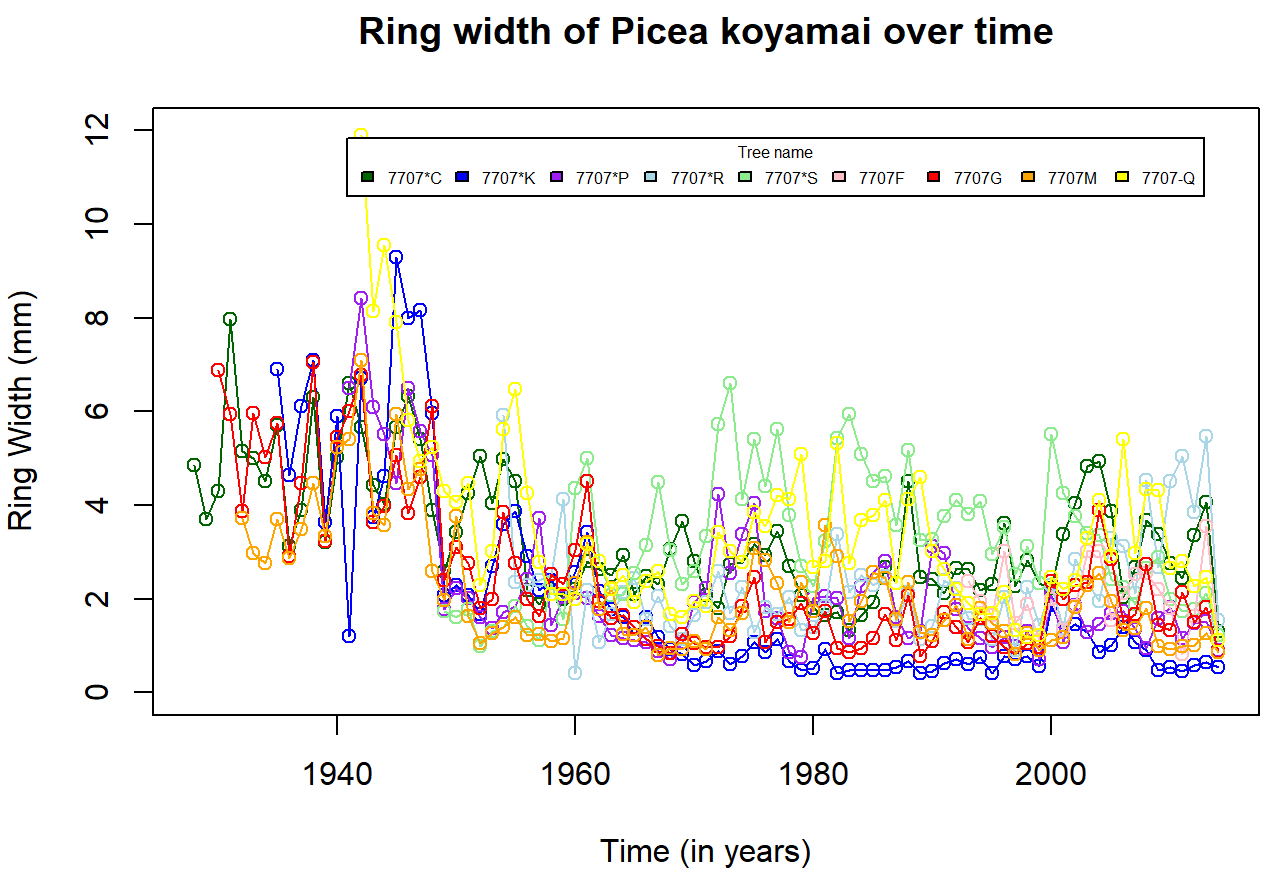
```
korai1 <- subset(gymno_data_long, TreeName == "503-40*B")


plot(korai1$Year, y = korai1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", ylal

legend("topright", inset = .05, title = "Tree name", c("503-40*B"), fill = c("darkgreen"), horiz
```



Picea koyamai

```
koya1 <- subset(gymno_data_long, TreeName == "7707*C")

koya2 <- subset(gymno_data_long, TreeName == "7707*K")

koya3 <- subset(gymno_data_long, TreeName == "7707*P")

koya4 <- subset(gymno_data_long, TreeName == "7707*R")

koya5 <- subset(gymno_data_long, TreeName == "7707*S")

koya6 <- subset(gymno_data_long, TreeName == "7707F")

koya7 <- subset(gymno_data_long, TreeName == "7707G")

koya8 <- subset(gymno_data_long, TreeName == "7707M")

koya9 <- subset(gymno_data_long, TreeName == "7707-Q")


plot(koya1$Year, y = koya1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", ylab =
lines(koya2$Year, y = koya2$Ringwidth, type = "o", col = "blue")
```

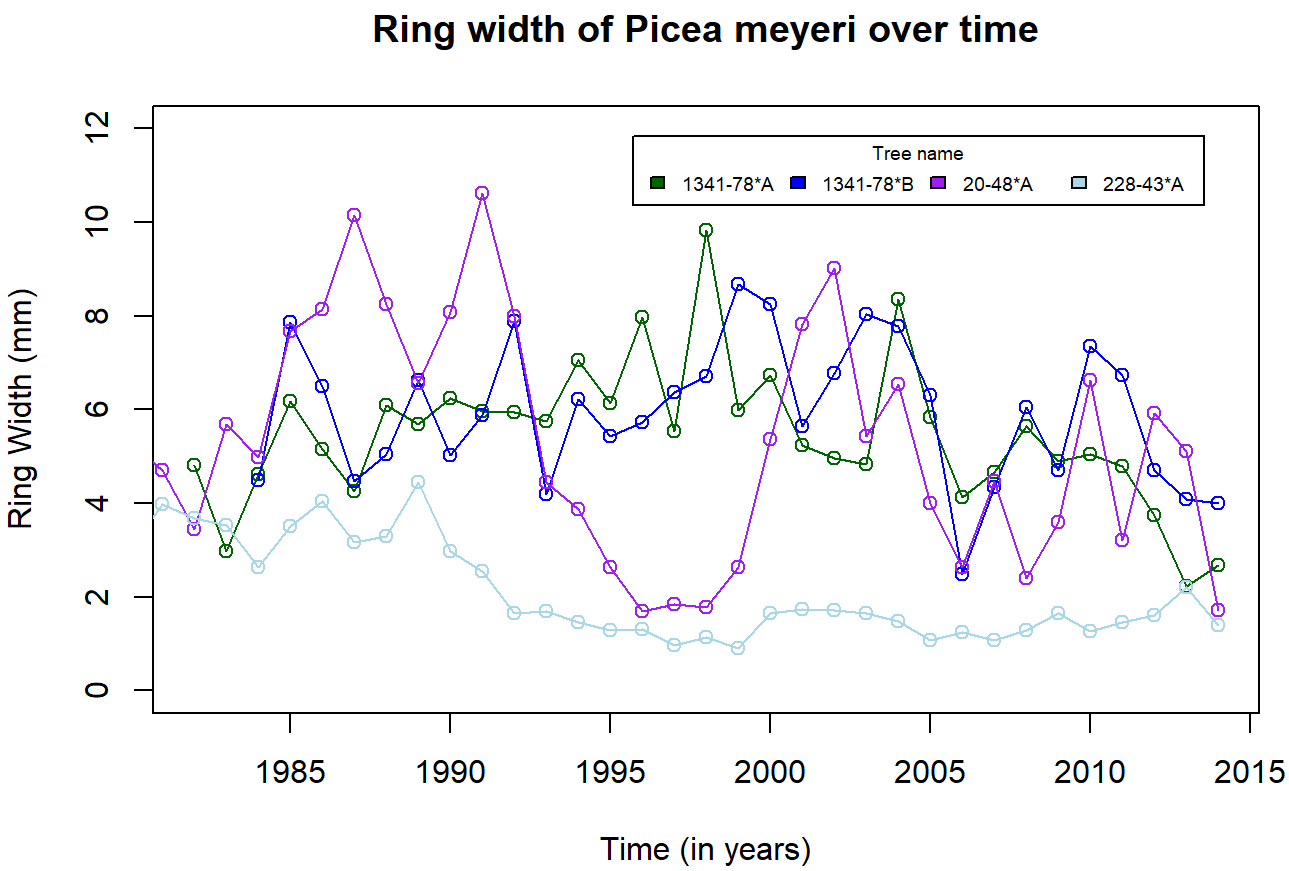
```
lines(koya3$Year, y = koya3$Ringwidth, type = "o", col = "purple")
lines(koya4$Year, y = koya4$Ringwidth, type = "o", col = "lightblue")
lines(koya5$Year, y = koya5$Ringwidth, type = "o", col = "lightgreen")
lines(koya6$Year, y = koya6$Ringwidth, type = "o", col = "pink")
lines(koya7$Year, y = koya7$Ringwidth, type = "o", col = "red")
lines(koya8$Year, y = koya8$Ringwidth, type = "o", col = "orange")
lines(koya9$Year, y = koya9$Ringwidth, type = "o", col = "yellow")

legend("topright", inset = .05, title = "Tree name", c("7707*C", "7707*K", "7707*P", "7707*R", "7
```



Picea meyeri

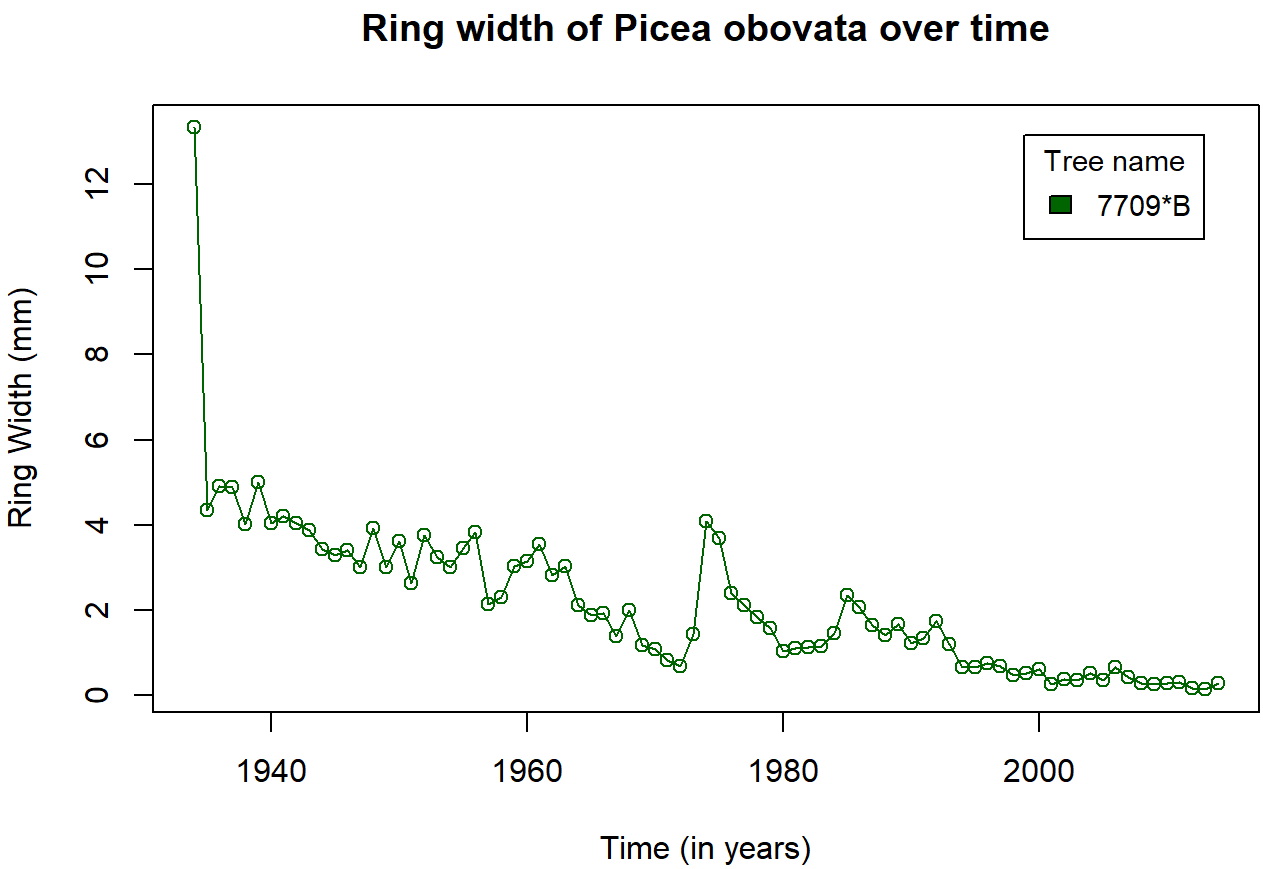
```
meye1 <- subset(gymno_data_long, TreeName == "1341-78*A")

meye2 <- subset(gymno_data_long, TreeName == "1341-78*B")

meye3 <- subset(gymno_data_long, TreeName == "20-48*A")

meye4 <- subset(gymno_data_long, TreeName == "228-43*A")


plot(meye1$Year, y = meye1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", ylab
```

```
lines(meye2$Year, y = meye2$Ringwidth, type = "o", col = "blue")
lines(meye3$Year, y = meye3$Ringwidth, type = "o", col = "purple")
lines(meye4$Year, y = meye4$Ringwidth, type = "o", col = "lightblue")

legend("topright", inset = .05, title = "Tree name", c("1341-78*A", "1341-78*B", "20-48*A", "228-
```



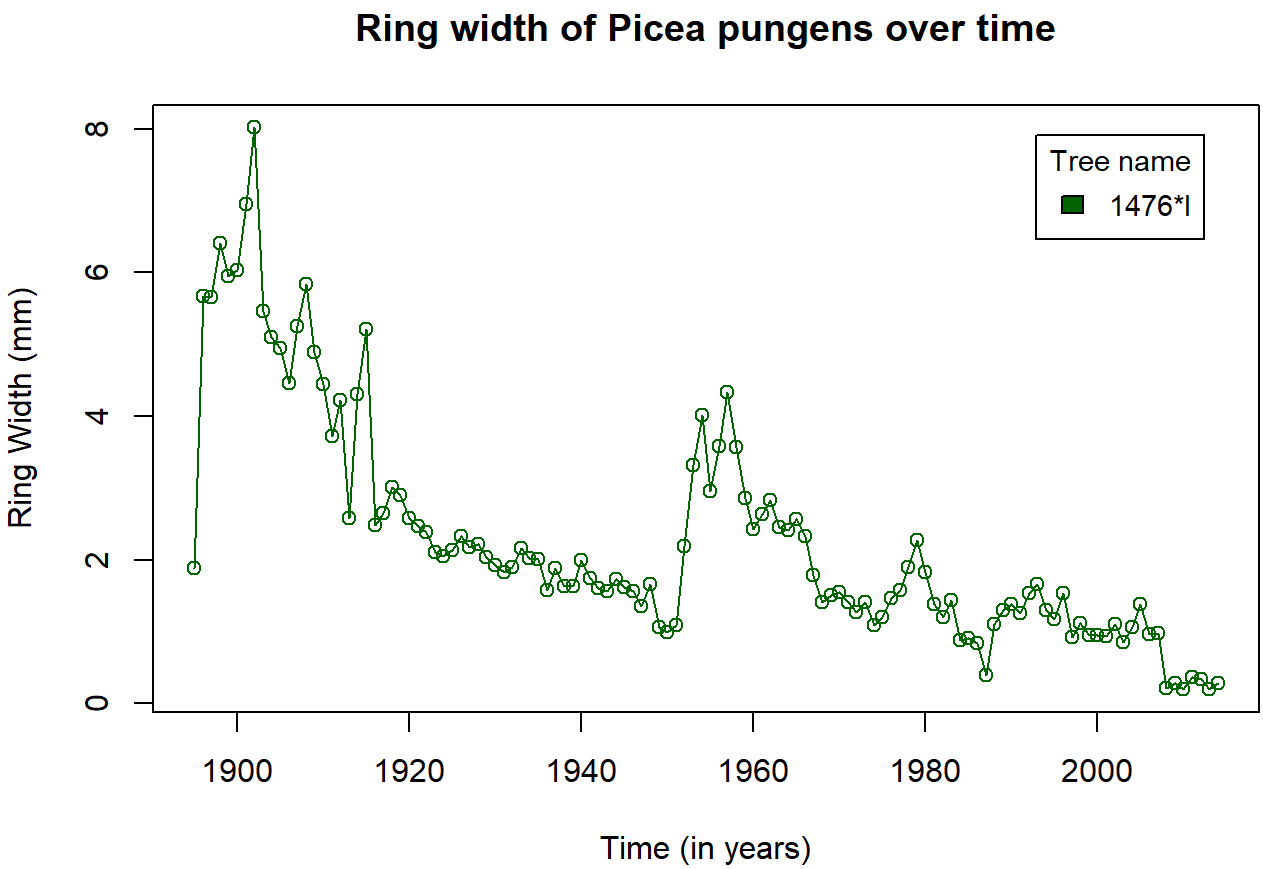
Picea obovata

```
obovata <- subset(gymno_data_long, TreeName == "7709*B")


plot(obovata$Year, y = obovata$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", y

legend("topright", inset = .05, title = "Tree name", c("7709*B"), fill = c("darkgreen"), horiz =
```



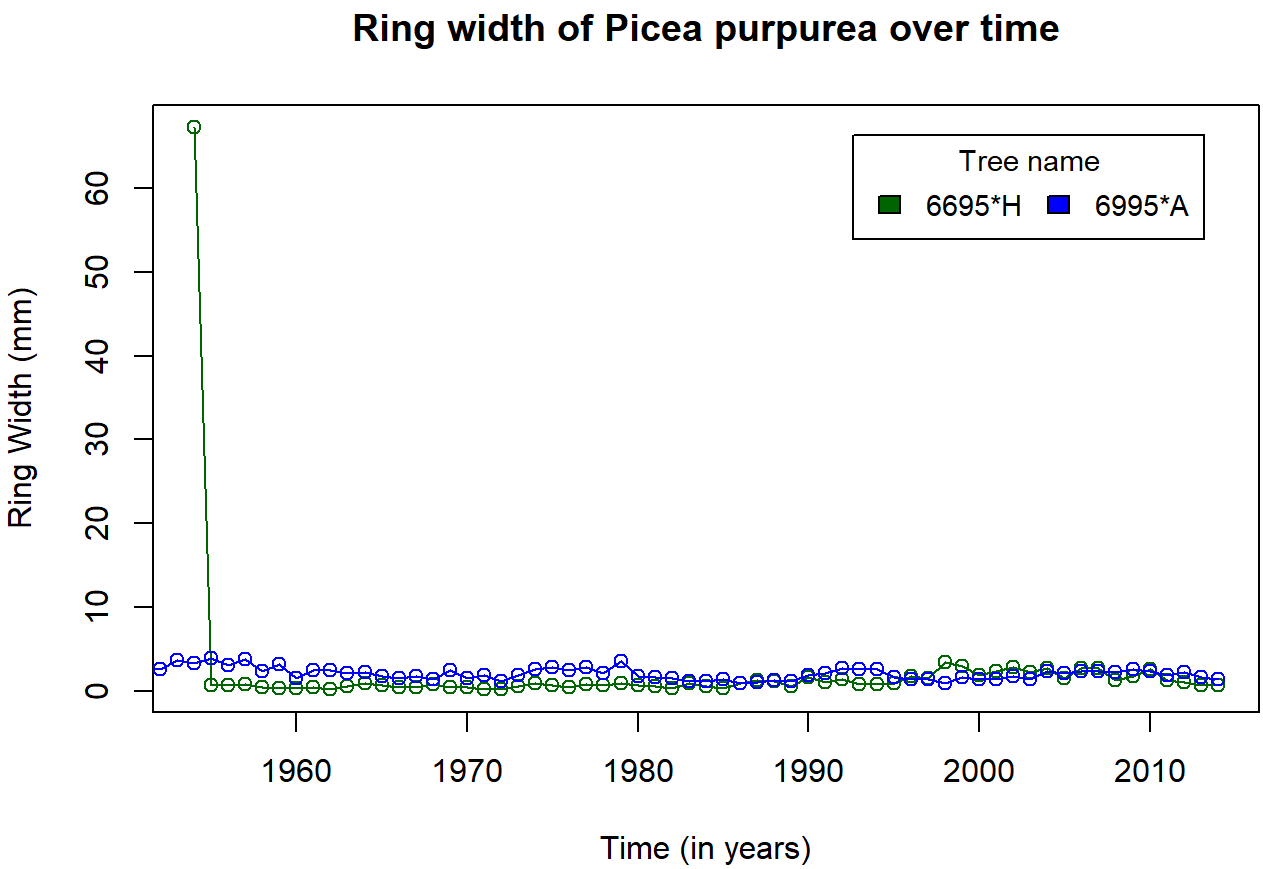
Picea pungens

```
pungens <- subset(gymno_data_long, TreeName == "1476*I")


plot(pungens$Year, y = pungens$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", y

legend("topright", inset = .05, title = "Tree name", c("1476*I"), fill = c("darkgreen"), horiz =
```



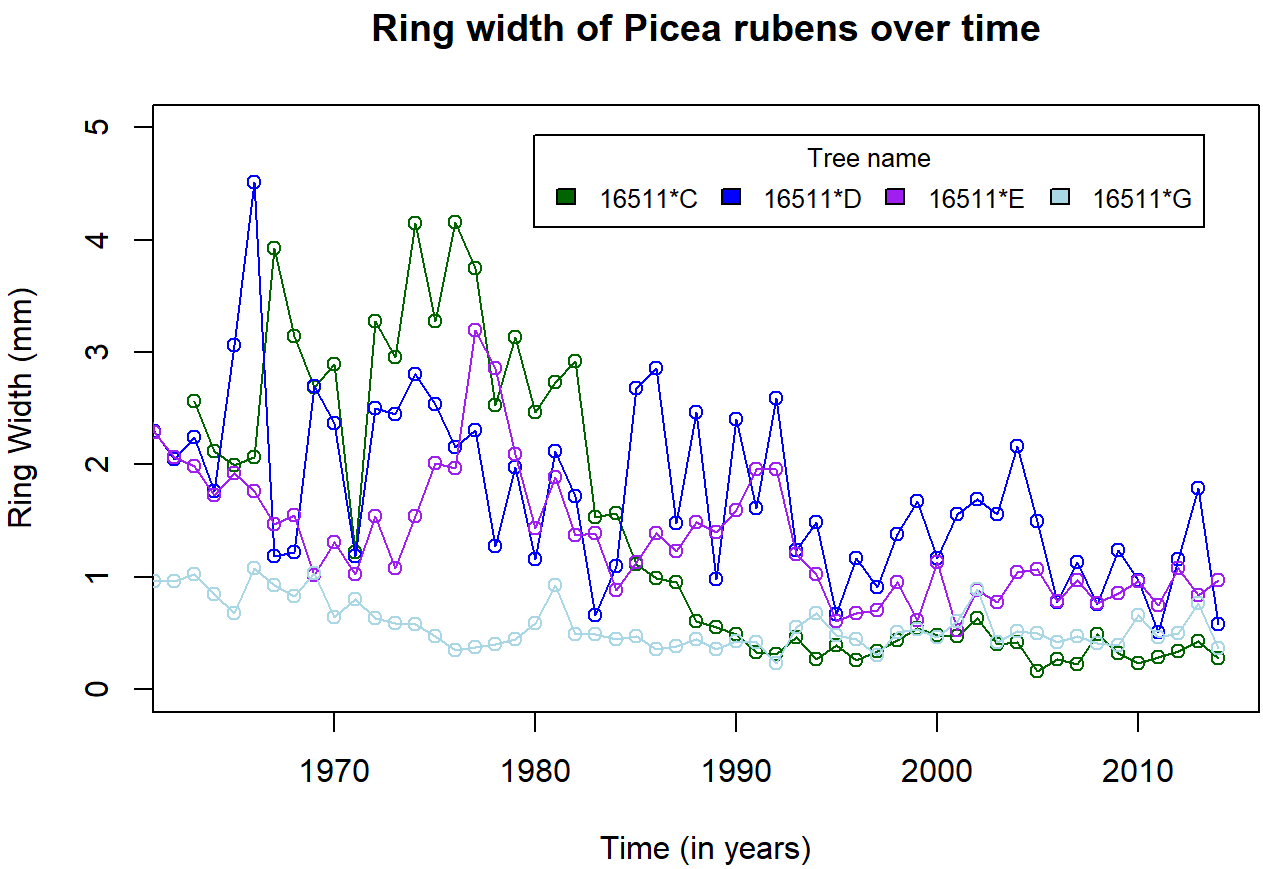
Picea purpurea

```
purpurea1 <- subset(gymno_data_long, TreeName == "6695*H")

purpurea2 <- subset(gymno_data_long, TreeName == "6995*A")


plot(purpurea1$Year, y = purpurea1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)
lines(purpurea2$Year, y = purpurea2$Ringwidth, type = "o", col = "blue")


legend("topright", inset = .05, title = "Tree name", c("6695*H", "6995*A"), fill = c("darkgreen",
```



Picea rubens

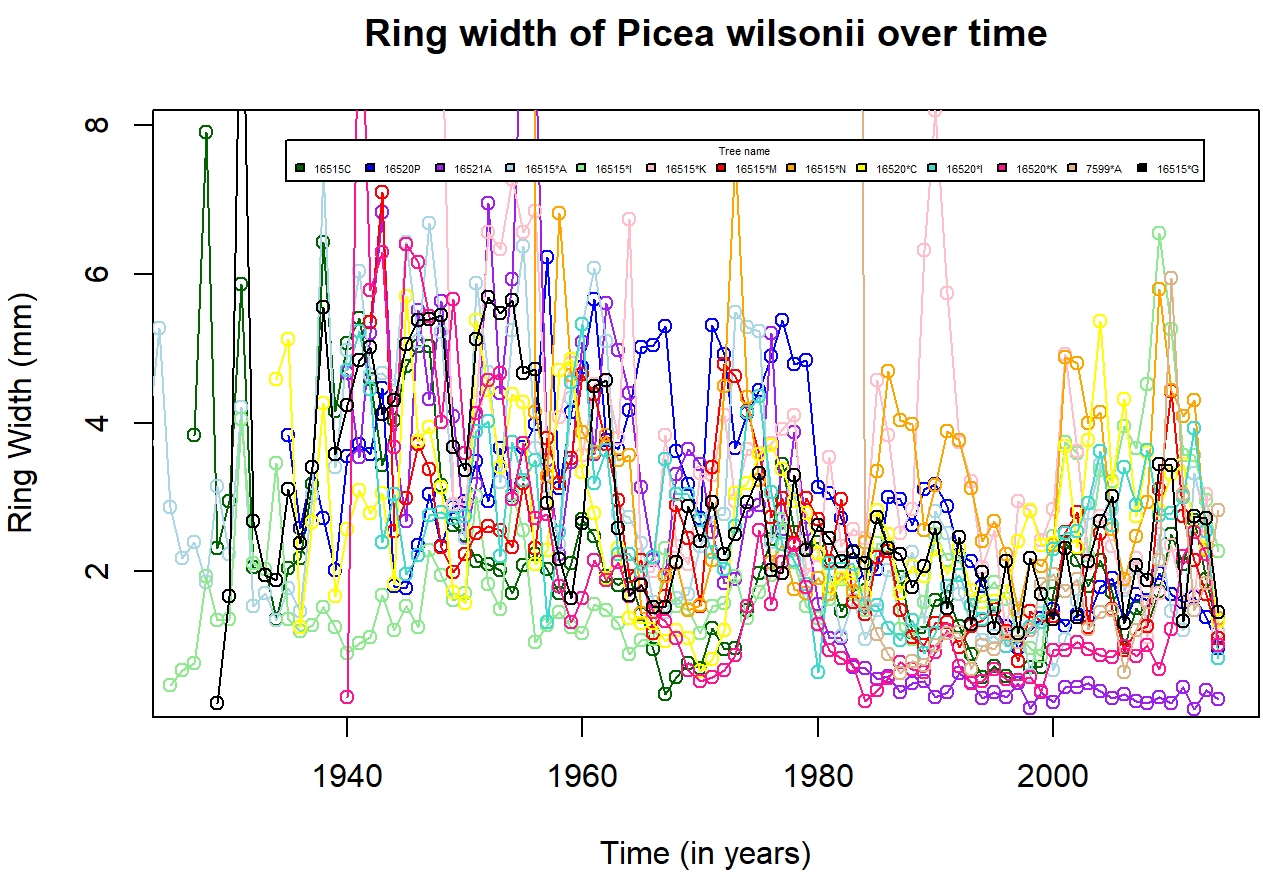
```
rubens1 <- subset(gymno_data_long, TreeName == "16511*C")

rubens2 <- subset(gymno_data_long, TreeName == "16511*D")

rubens3 <- subset(gymno_data_long, TreeName == "16511*E")

rubens4 <- subset(gymno_data_long, TreeName == "16511*G")


plot(rubens1$Year, y = rubens1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", y
lines(rubens2$Year, y = rubens2$Ringwidth, type = "o", col = "blue")
lines(rubens3$Year, y = rubens3$Ringwidth, type = "o", col = "purple")
lines(rubens4$Year, y = rubens4$Ringwidth, type = "o", col = "lightblue")

legend("topright", inset = .05, title = "Tree name", c("16511*C", "16511*D", "16511*E", "16511*G"
```



Picea wilsonii

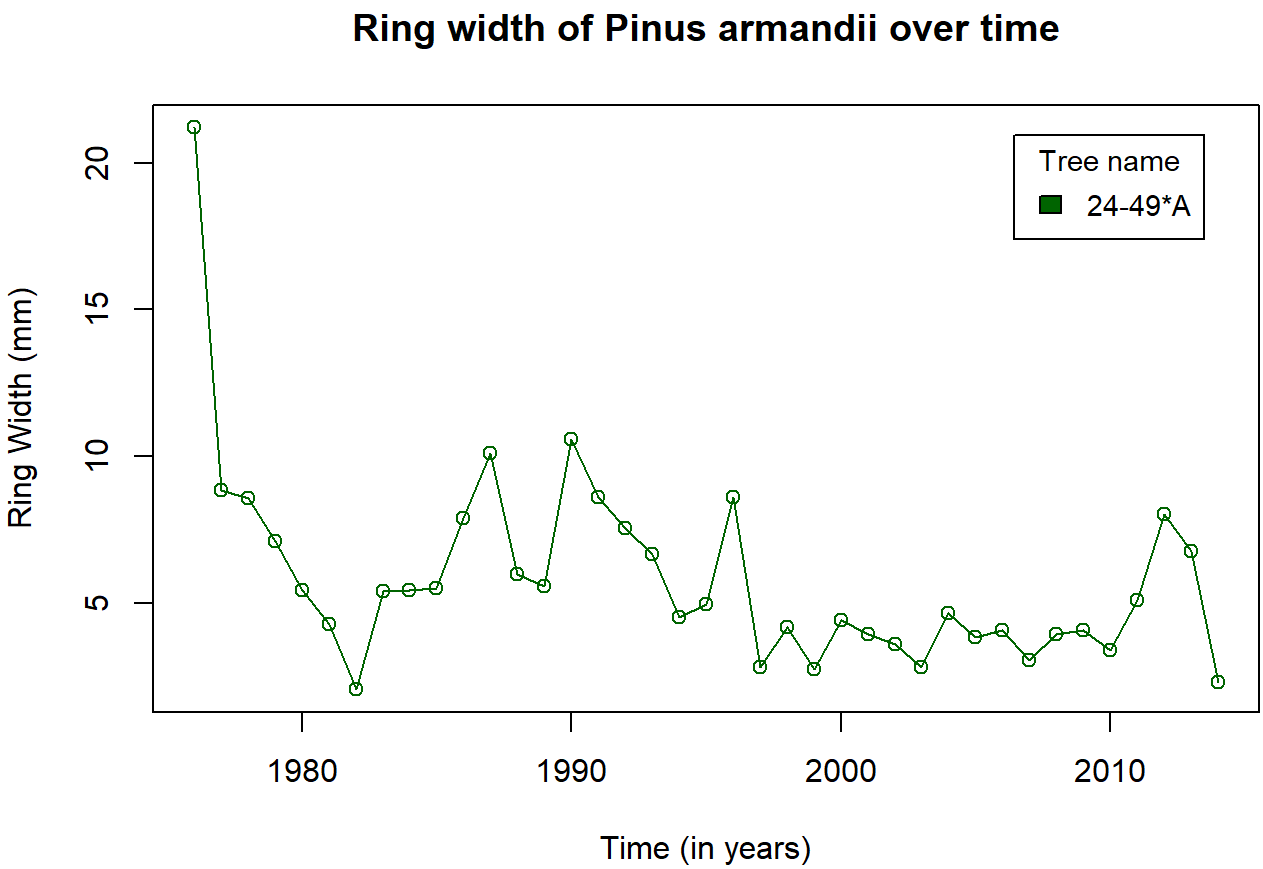
```
wils1 <- subset(gymno_data_long, TreeName == "16515C")

wils2 <- subset(gymno_data_long, TreeName == "16520P")

wils3 <- subset(gymno_data_long, TreeName == "16521A")

wils4 <- subset(gymno_data_long, TreeName == "16515*A")

wils5 <- subset(gymno_data_long, TreeName == "16515*I")

wils6 <- subset(gymno_data_long, TreeName == "16515*K")

wils7 <- subset(gymno_data_long, TreeName == "16515*M")

wils8 <- subset(gymno_data_long, TreeName == "16515*N")

wils9 <- subset(gymno_data_long, TreeName == "16520*C")

wils10 <- subset(gymno_data_long, TreeName == "16520*I")

wils11 <- subset(gymno_data_long, TreeName == "16520*K")
```

```
wils12 <- subset(gymno_data_long, TreeName == "7599*A")

wils13 <- subset(gymno_data_long, TreeName == "16515*G")


plot(wils1$Year, y = wils1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", ylab =
lines(wils2$Year, y = wils2$Ringwidth, type = "o", col = "blue")
lines(wils3$Year, y = wils3$Ringwidth, type = "o", col = "purple")
lines(wils4$Year, y = wils4$Ringwidth, type = "o", col = "lightblue")
lines(wils5$Year, y = wils5$Ringwidth, type = "o", col = "lightgreen")
lines(wils6$Year, y = wils6$Ringwidth, type = "o", col = "pink")
lines(wils7$Year, y = wils7$Ringwidth, type = "o", col = "red")
lines(wils8$Year, y = wils8$Ringwidth, type = "o", col = "orange")
lines(wils9$Year, y = wils9$Ringwidth, type = "o", col = "yellow")
lines(wils10$Year, y = wils10$Ringwidth, type = "o", col = "turquoise")
lines(wils11$Year, y = wils11$Ringwidth, type = "o", col = "deeppink")
lines(wils12$Year, y = wils12$Ringwidth, type = "o", col = "burlywood")
lines(wils13$Year, y = wils13$Ringwidth, type = "o", col = "black")

legend("topright", inset = .05, title = "Tree name", c("16515C", "16520P", "16521A", "16515*A", "
```



Pinus armandii

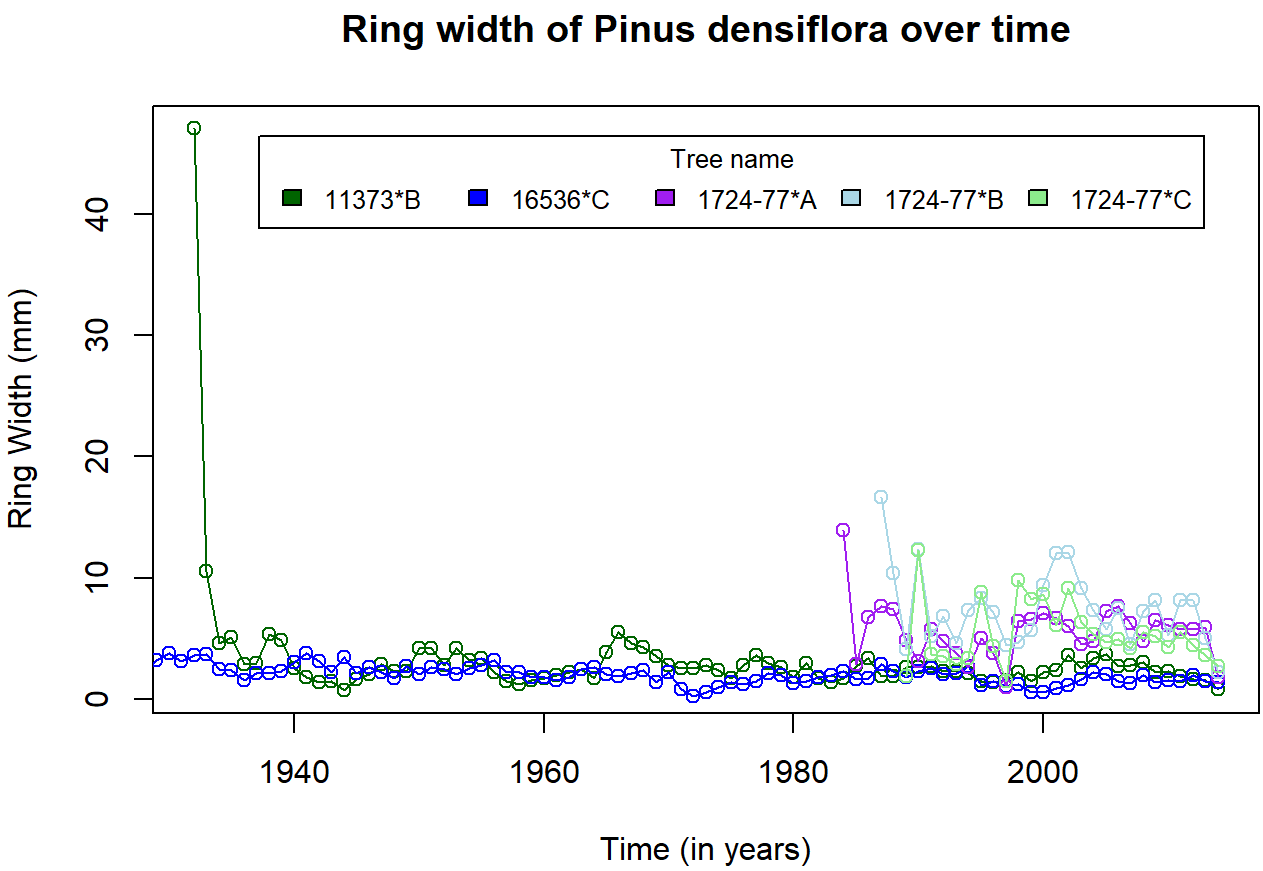
```
armandii <- subset(gymno_data_long, TreeName == "24-49*A")


plot(armandii$Year, y = armandii$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)",

legend("topright", inset = .05, title = "Tree name", c("24-49*A"), fill = c("darkgreen"), horiz =
```



Pinus densiflora

```
densi1 <- subset(gymno_data_long, TreeName == "11373*B")

densi2 <- subset(gymno_data_long, TreeName == "16536*C")

densi3 <- subset(gymno_data_long, TreeName == "1724-77*A")

densi4 <- subset(gymno_data_long, TreeName == "1724-77*B")

densi5 <- subset(gymno_data_long, TreeName == "1724-77*C")
```

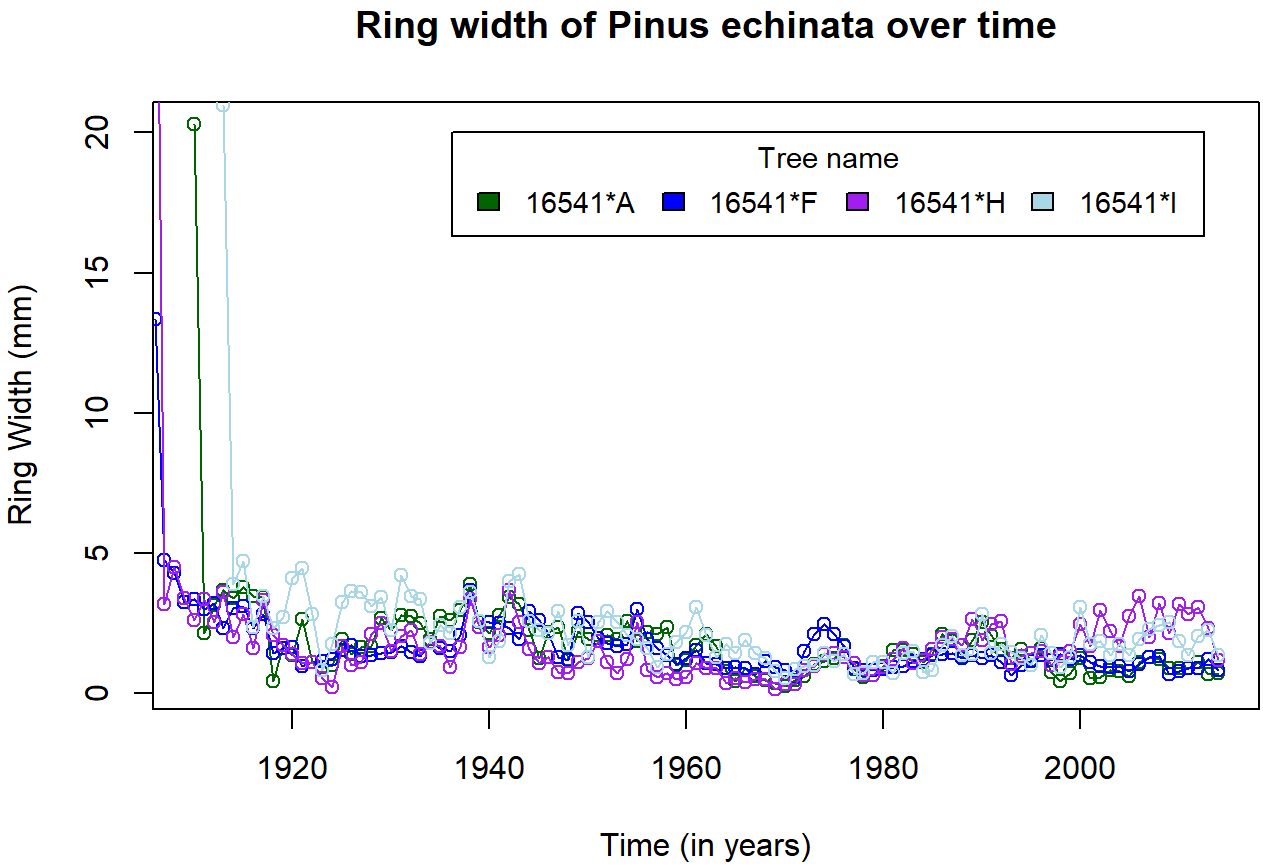
```
plot(densi1$Year, y = densi1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", ylal
lines(densi2$Year, y = densi2$Ringwidth, type = "o", col = "blue")
lines(densi3$Year, y = densi3$Ringwidth, type = "o", col = "purple")
lines(densi4$Year, y = densi4$Ringwidth, type = "o", col = "lightblue")
lines(densi5$Year, y = densi5$Ringwidth, type = "o", col = "lightgreen")

legend("topright", inset = .05, title = "Tree name", c("11373*B", "16536*C", "1724-77*A", "1724-7
```



Pinus echinata

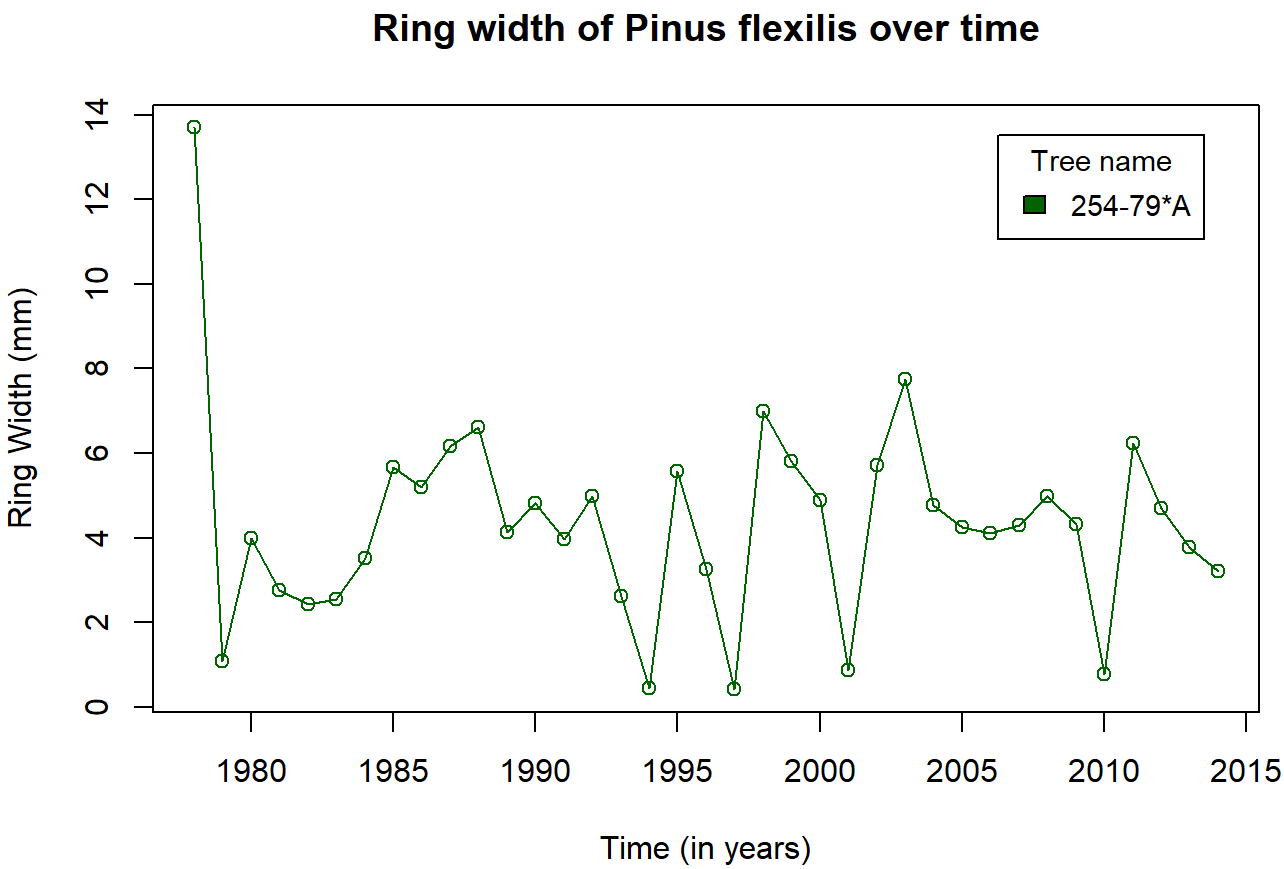
```
echin1 <- subset(gymno_data_long, TreeName == "16541*A")

echin2 <- subset(gymno_data_long, TreeName == "16541*F")

echin3 <- subset(gymno_data_long, TreeName == "16541*H")

echin4 <- subset(gymno_data_long, TreeName == "16541*I")


plot(echin1$Year, y = echin1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", ylal
lines(echin2$Year, y = echin2$Ringwidth, type = "o", col = "blue")
```

```
lines(echin3$Year, y = echin3$Ringwidth, type = "o", col = "purple")
lines(echin4$Year, y = echin4$Ringwidth, type = "o", col = "lightblue")

legend("topright", inset = .05, title = "Tree name", c("16541*A", "16541*F", "16541*H", "16541*I"
```



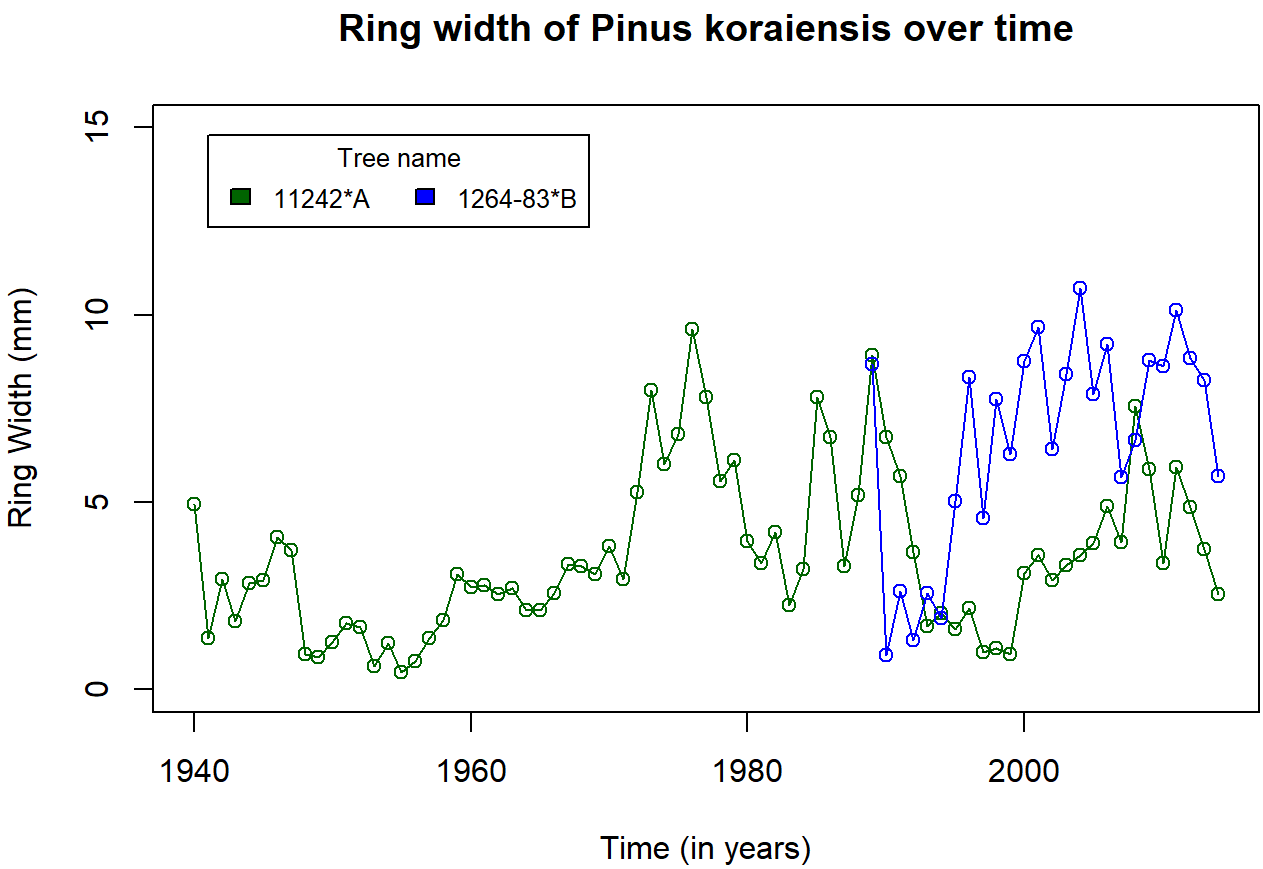
Pinus flexilis

```
flexilis <- subset(gymno_data_long, TreeName == "254-79*A")


plot(flexilis$Year, y = flexilis$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)",

legend("topright", inset = .05, title = "Tree name", c("254-79*A"), fill = c("darkgreen"), horiz
```



Pinus koraiensis

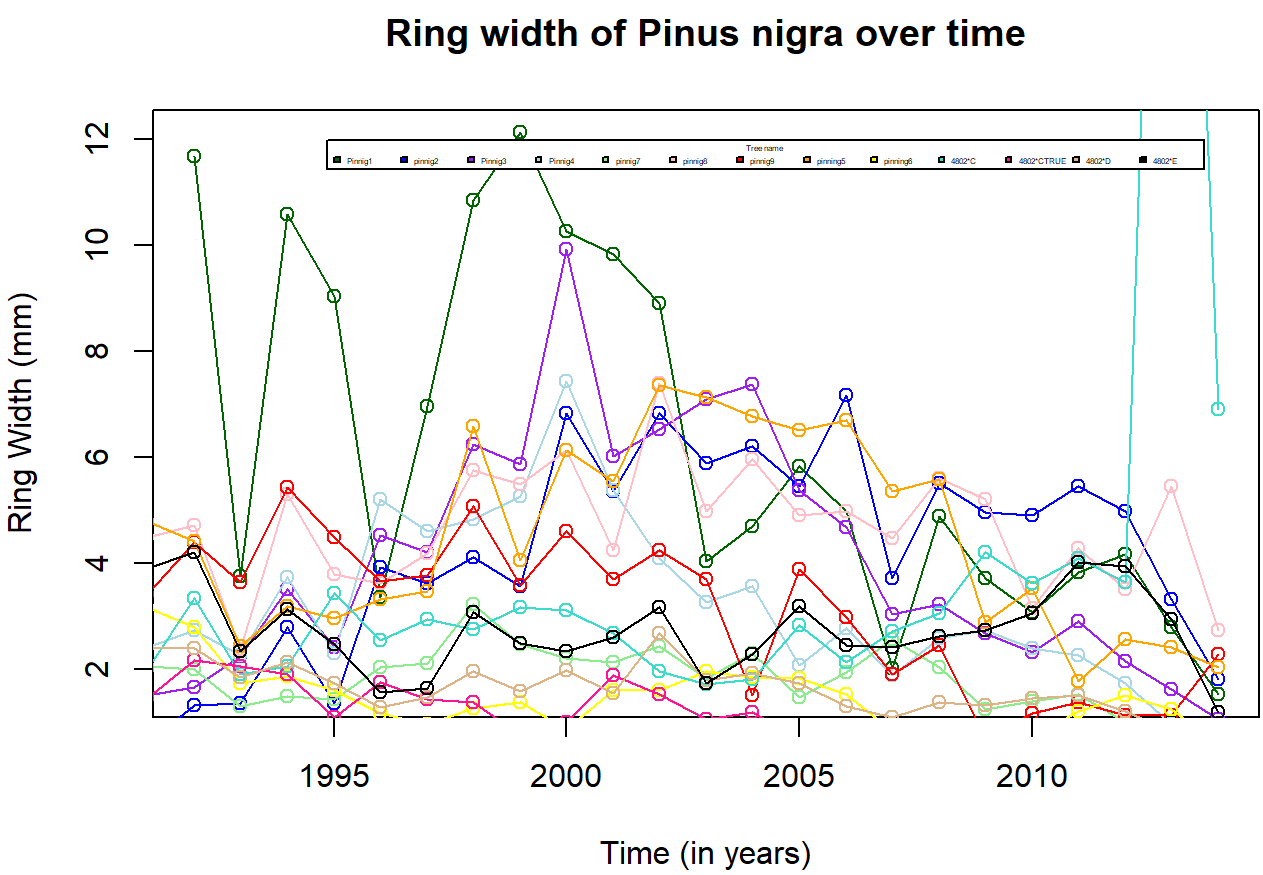
```
pkorai1 <- subset(gymno_data_long, TreeName == "11242*A")

pkorai2 <- subset(gymno_data_long, TreeName == "1264-83*B")


plot(pkorai1$Year, y = pkorai1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", y
lines(pkorai2$Year, y = pkorai2$Ringwidth, type = "o", col = "blue")

legend("topleft", inset = .05, title = "Tree name", c("11242*A", "1264-83*B"), fill = c("darkgree
```



Pinus nigra

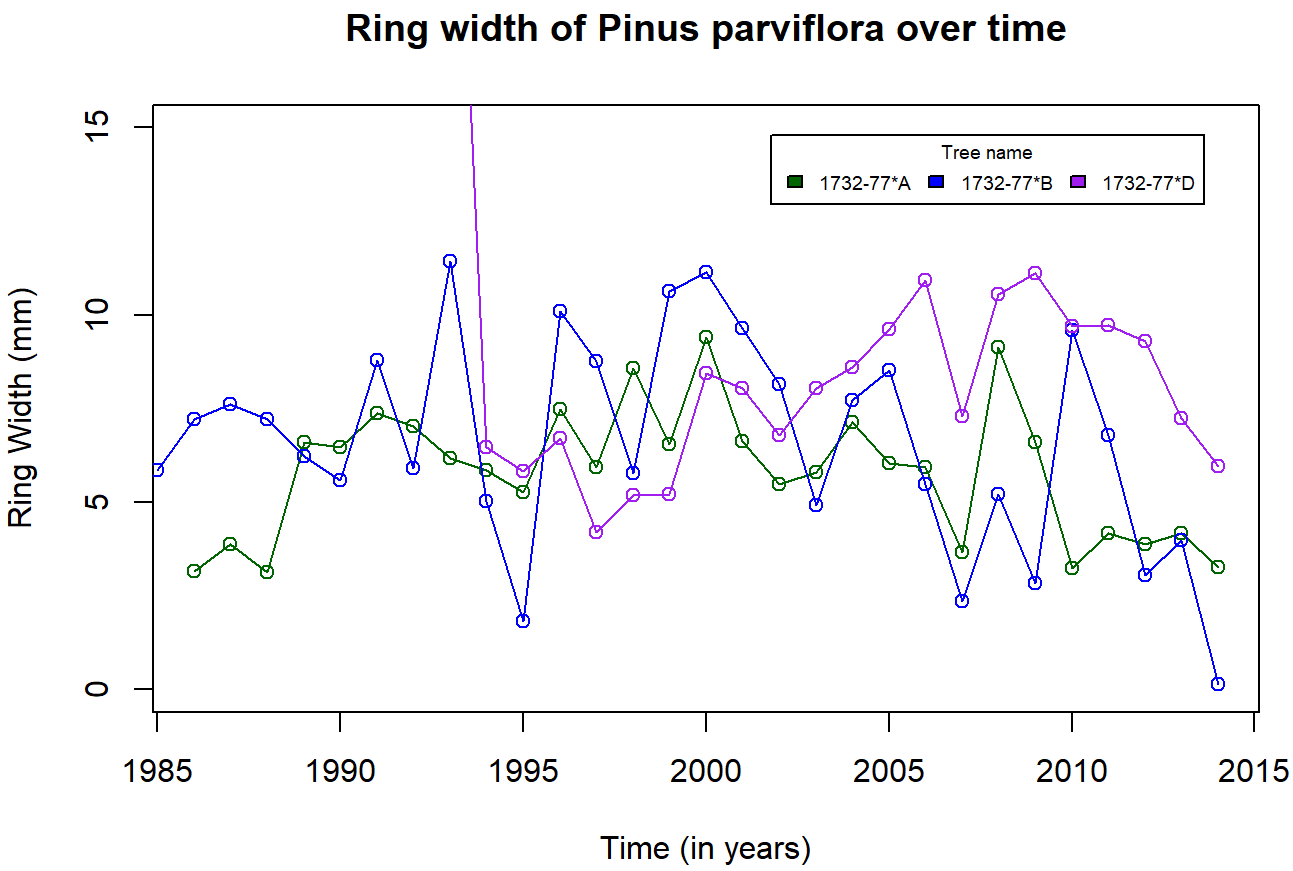
```
nigra1 <- subset(gymno_data_long, TreeName == "Pinnig1")

nigra2 <- subset(gymno_data_long, TreeName == "pinnig2")

nigra3 <- subset(gymno_data_long, TreeName == "Pinnig3")

nigra4 <- subset(gymno_data_long, TreeName == "Pinnig4")

nigra5 <- subset(gymno_data_long, TreeName == "pinnig7")

nigra6 <- subset(gymno_data_long, TreeName == "pinnig8")

nigra7 <- subset(gymno_data_long, TreeName == "pinnig9")

nigra8 <- subset(gymno_data_long, TreeName == "pinning5")

nigra9 <- subset(gymno_data_long, TreeName == "pinning6")

nigra10 <- subset(gymno_data_long, TreeName == "4802*C")

nigra11 <- subset(gymno_data_long, TreeName == "4802*CTRUE")
```

```
nigra12 <- subset(gymno_data_long, TreeName == "4802*D")

nigra13 <- subset(gymno_data_long, TreeName == "4802*E")


plot(nigra1$Year, y = nigra1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", ylal
lines(nigra2$Year, y = nigra2$Ringwidth, type = "o", col = "blue")
lines(nigra3$Year, y = nigra3$Ringwidth, type = "o", col = "purple")
lines(nigra4$Year, y = nigra4$Ringwidth, type = "o", col = "lightblue")
lines(nigra5$Year, y = nigra5$Ringwidth, type = "o", col = "lightgreen")
lines(nigra6$Year, y = nigra6$Ringwidth, type = "o", col = "pink")
lines(nigra7$Year, y = nigra7$Ringwidth, type = "o", col = "red")
lines(nigra8$Year, y = nigra8$Ringwidth, type = "o", col = "orange")
lines(nigra9$Year, y = nigra9$Ringwidth, type = "o", col = "yellow")
lines(nigra10$Year, y = nigra10$Ringwidth, type = "o", col = "turquoise")
lines(nigra11$Year, y = nigra11$Ringwidth, type = "o", col = "deeppink")
lines(nigra12$Year, y = nigra12$Ringwidth, type = "o", col = "burlywood")
lines(nigra13$Year, y = nigra13$Ringwidth, type = "o", col = "black")

legend("topright", inset = .05, title = "Tree name", c("Pinnig1", "pinnig2", "Pinnig3", "Pinnig4"
```



Pinus parviflora

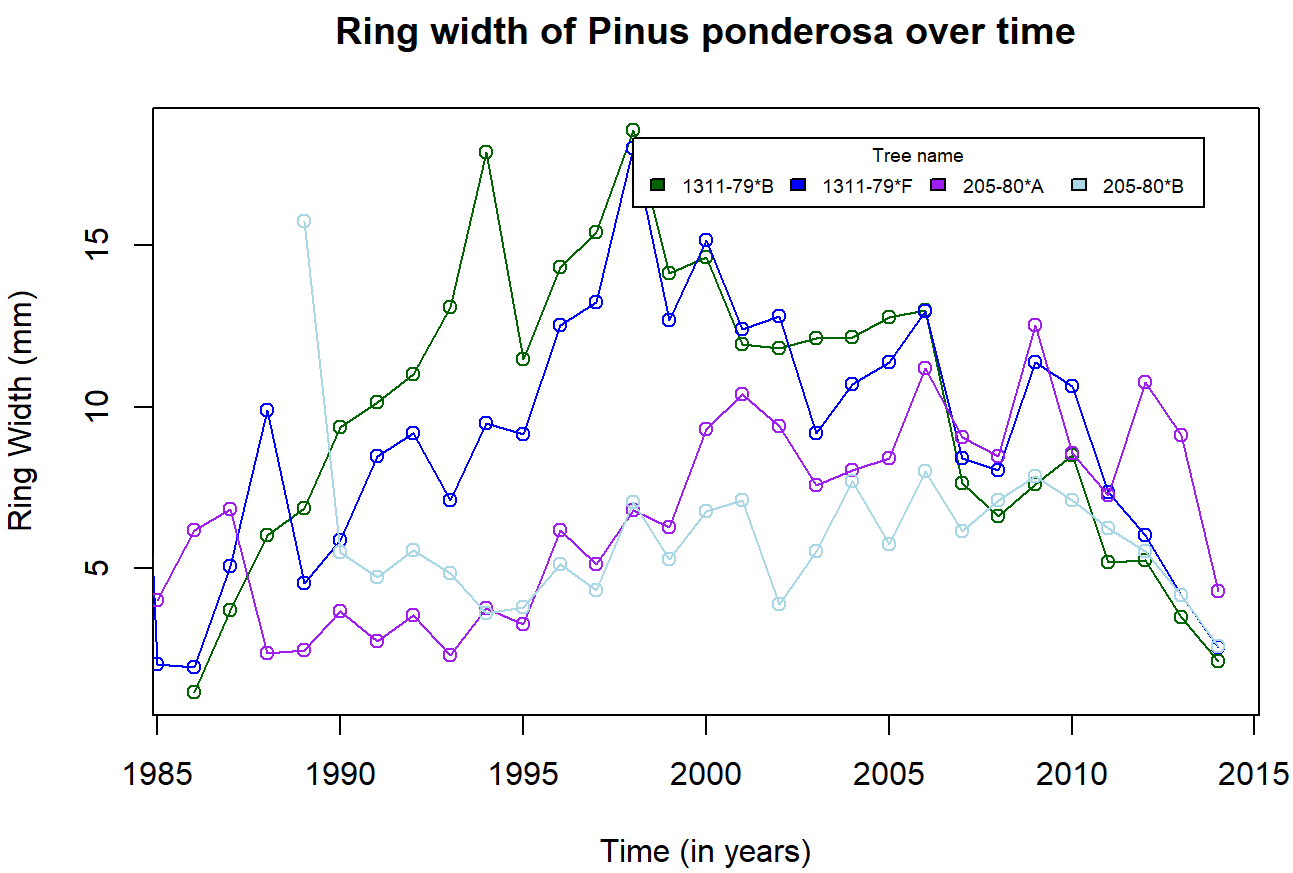
```
parvi1 <- subset(gymno_data_long, TreeName == "1732-77*A")

parvi2 <- subset(gymno_data_long, TreeName == "1732-77*B")

parvi3 <- subset(gymno_data_long, TreeName == "1732-77*D")


plot(parvi1$Year, y = parvi1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", ylal
lines(parvi2$Year, y = parvi2$Ringwidth, type = "o", col = "blue")
lines(parvi3$Year, y = parvi3$Ringwidth, type = "o", col = "purple")

legend("topright", inset = .05, title = "Tree name", c("1732-77*A", "1732-77*B", "1732-77*D"), fi
```



Pinus ponderosa

```
pond1 <- subset(gymno_data_long, TreeName == "1311-79*B")

pond2 <- subset(gymno_data_long, TreeName == "1311-79*F")
```

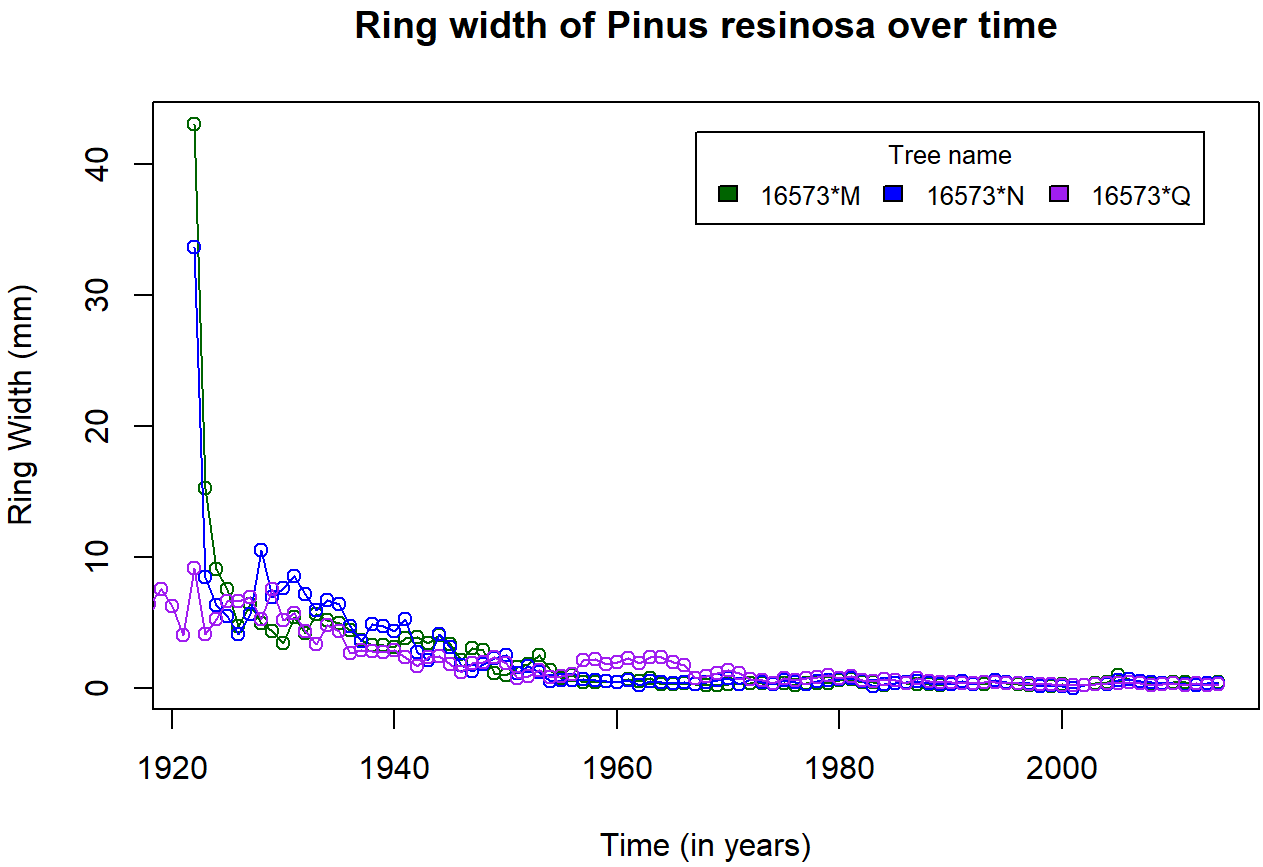
```
pond3 <- subset(gymno_data_long, TreeName == "205-80*A")

pond4 <- subset(gymno_data_long, TreeName == "205-80*B")


plot(pond1$Year, y = pond1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", ylab =
lines(pond2$Year, y = pond2$Ringwidth, type = "o", col = "blue")
lines(pond3$Year, y = pond3$Ringwidth, type = "o", col = "purple")
lines(pond4$Year, y = pond4$Ringwidth, type = "o", col = "lightblue")

legend("topright", inset = .05, title = "Tree name", c("1311-79*B", "1311-79*F", "205-80*A", "205
```



Pinus resinosa

```
resin1 <- subset(gymno_data_long, TreeName == "16573*M")

resin2 <- subset(gymno_data_long, TreeName == "16573*N")

resin3 <- subset(gymno_data_long, TreeName == "16573*Q")


plot(resin1$Year, y = resin1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", ylal
```

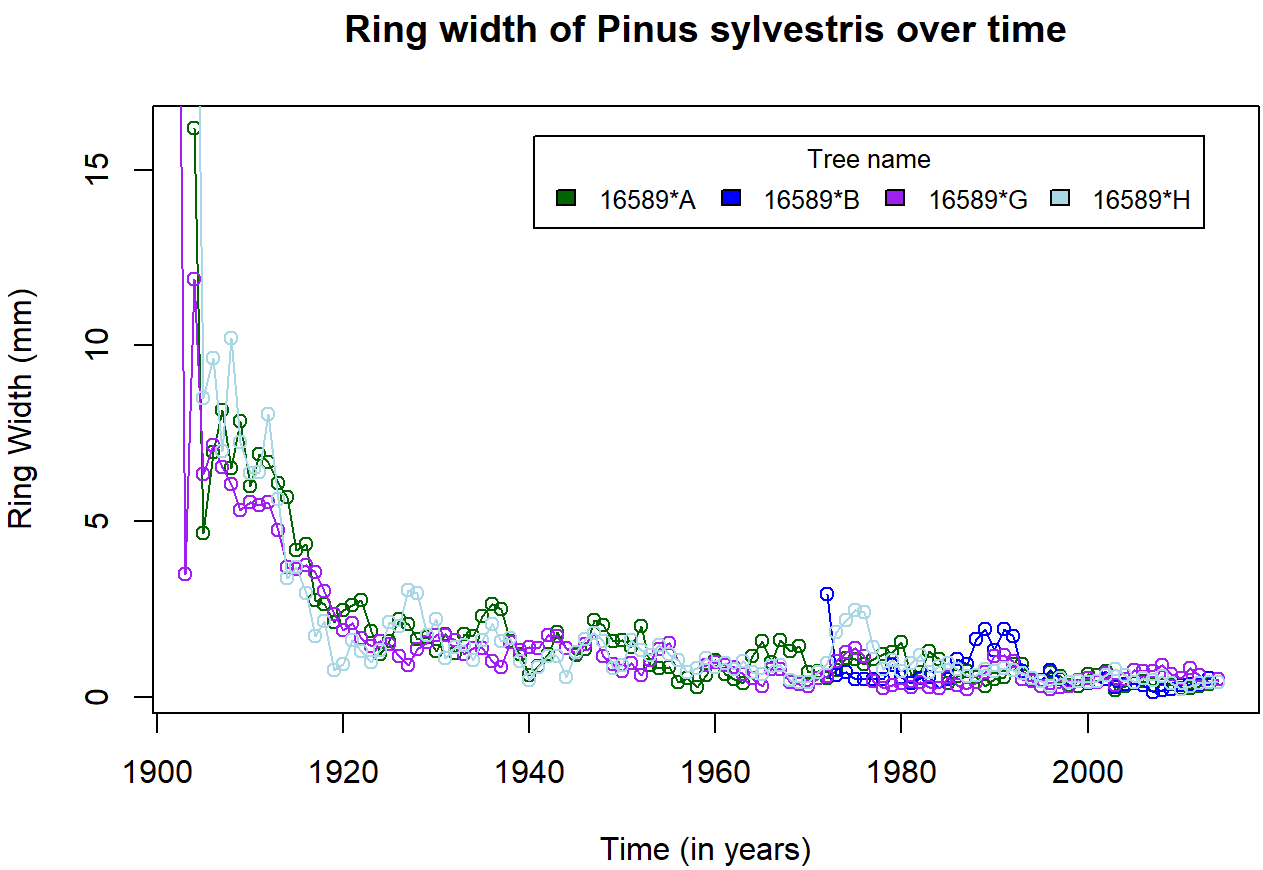
```
lines(resin2$Year, y = resin2$Ringwidth, type = "o", col = "blue")
lines(resin3$Year, y = resin3$Ringwidth, type = "o", col = "purple")

legend("topright", inset = .05, title = "Tree name", c("16573*M", "16573*N", "16573*Q"), fill = c
```



Pinus sylvestris

```
sylv1 <- subset(gymno_data_long, TreeName == "16589*A")

sylv2 <- subset(gymno_data_long, TreeName == "16589*B")

sylv3 <- subset(gymno_data_long, TreeName == "16589*G")

sylv4 <- subset(gymno_data_long, TreeName == "16589*H")


plot(sylv1$Year, y = sylv1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", ylab =
lines(sylv2$Year, y = sylv2$Ringwidth, type = "o", col = "blue")
lines(sylv3$Year, y = sylv3$Ringwidth, type = "o", col = "purple")
lines(sylv4$Year, y = sylv4$Ringwidth, type = "o", col = "lightblue")
```

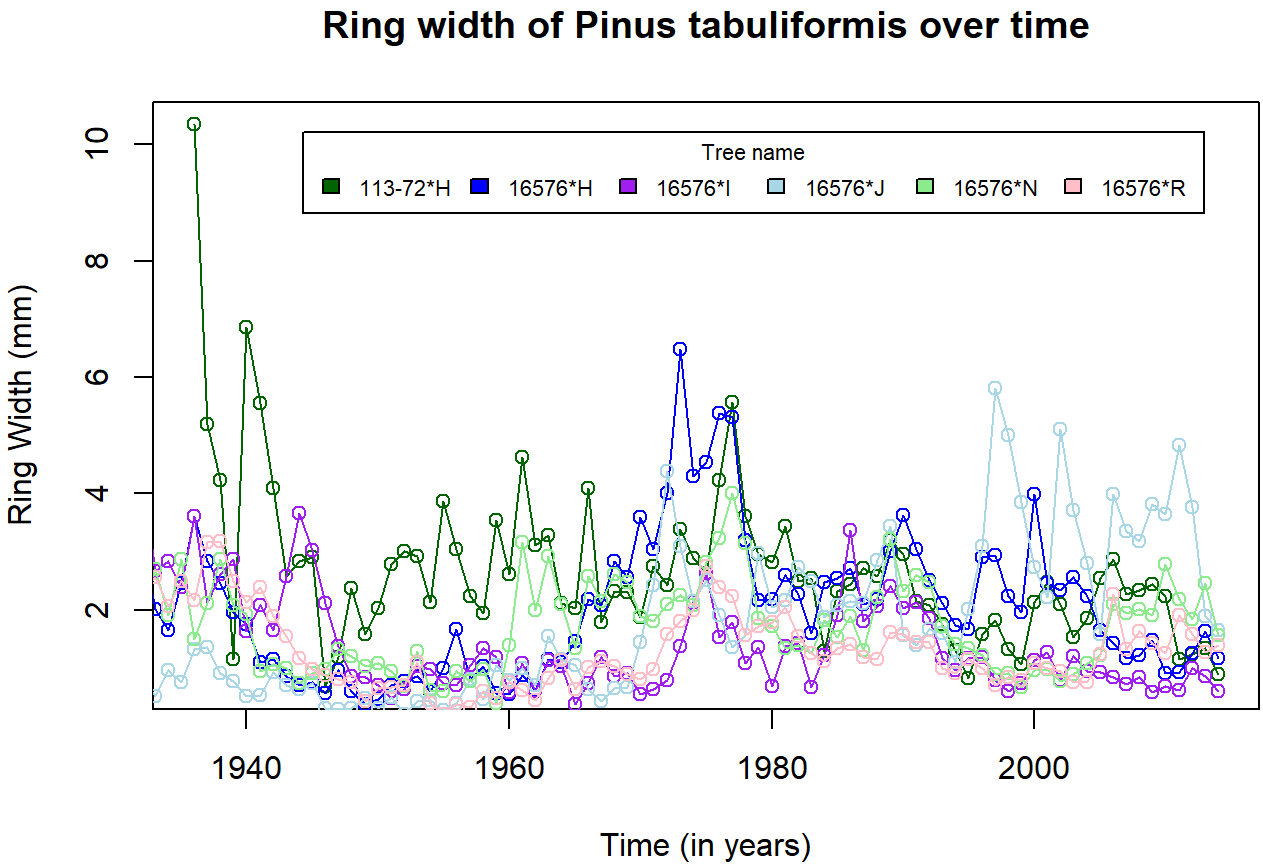
```
legend("topright", inset = .05, title = "Tree name", c("16589*A", "16589*B", "16589*G", "16589*H"
```



Pinus tabuliformis

```
tab1 <- subset(gymno_data_long, TreeName == "113-72*H")

tab2 <- subset(gymno_data_long, TreeName == "16576*H")

tab3 <- subset(gymno_data_long, TreeName == "16576*I")

tab4 <- subset(gymno_data_long, TreeName == "16576*J")

tab5 <- subset(gymno_data_long, TreeName == "16576*N")

tab6 <- subset(gymno_data_long, TreeName == "16576*R")


plot(tab1$Year, y = tab1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", ylab =
lines(tab2$Year, y = tab2$Ringwidth, type = "o", col = "blue")
lines(tab3$Year, y = tab3$Ringwidth, type = "o", col = "purple")
lines(tab4$Year, y = tab4$Ringwidth, type = "o", col = "lightblue")
```

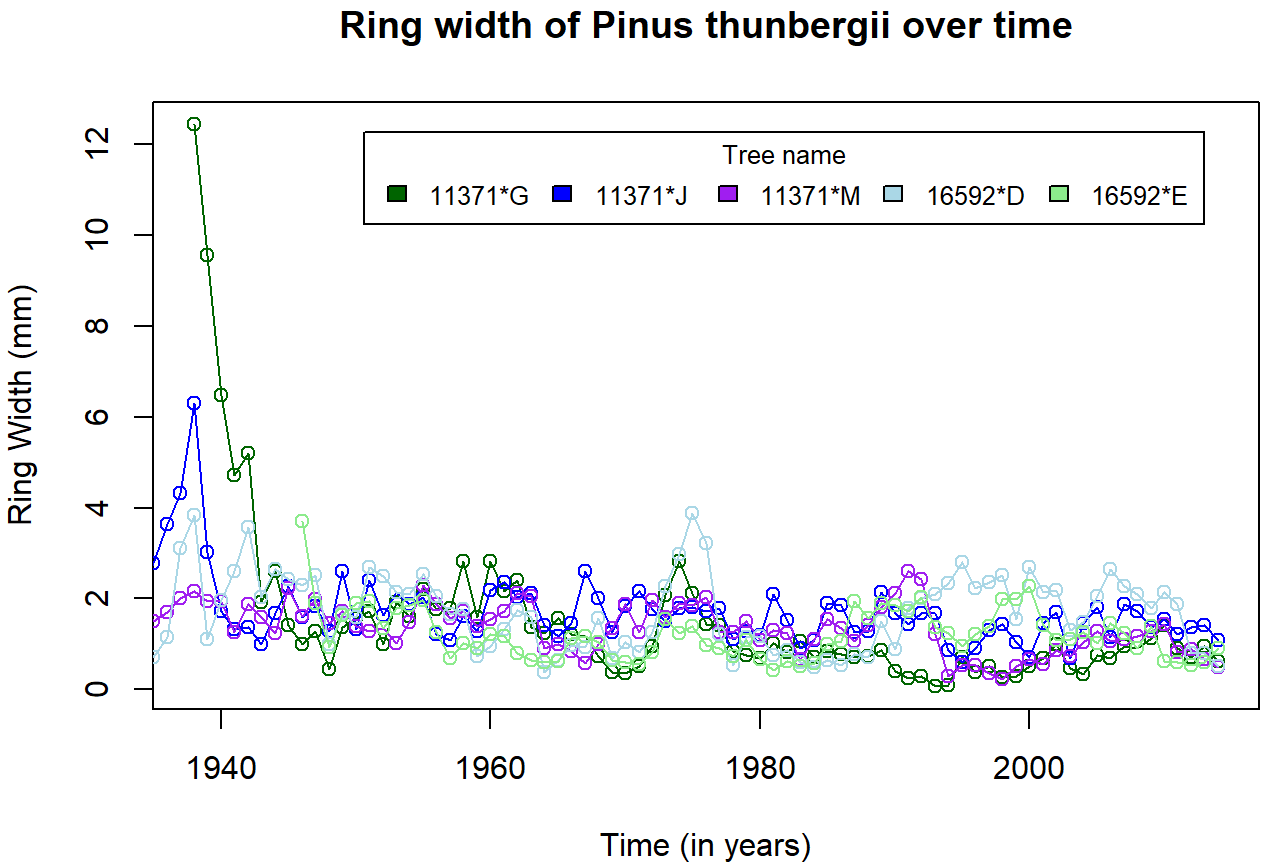
```
lines(tab5$Year, y = tab5$Ringwidth, type = "o", col = "lightgreen")
lines(tab6$Year, y = tab6$Ringwidth, type = "o", col = "pink")

legend("topright", inset = .05, title = "Tree name", c("113-72*H", "16576*H", "16576*I", "16576*J
```



Pinus thunbergii

```
thun1 <- subset(gymno_data_long, TreeName == "11371*G")

thun2 <- subset(gymno_data_long, TreeName == "11371*J")

thun3 <- subset(gymno_data_long, TreeName == "11371*M")

thun4 <- subset(gymno_data_long, TreeName == "16592*D")

thun5 <- subset(gymno_data_long, TreeName == "16592*E")


plot(thun1$Year, y = thun1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", ylab
lines(thun2$Year, y = thun2$Ringwidth, type = "o", col = "blue")
lines(thun3$Year, y = thun3$Ringwidth, type = "o", col = "purple")
lines(thun4$Year, y = thun4$Ringwidth, type = "o", col = "lightblue")
```

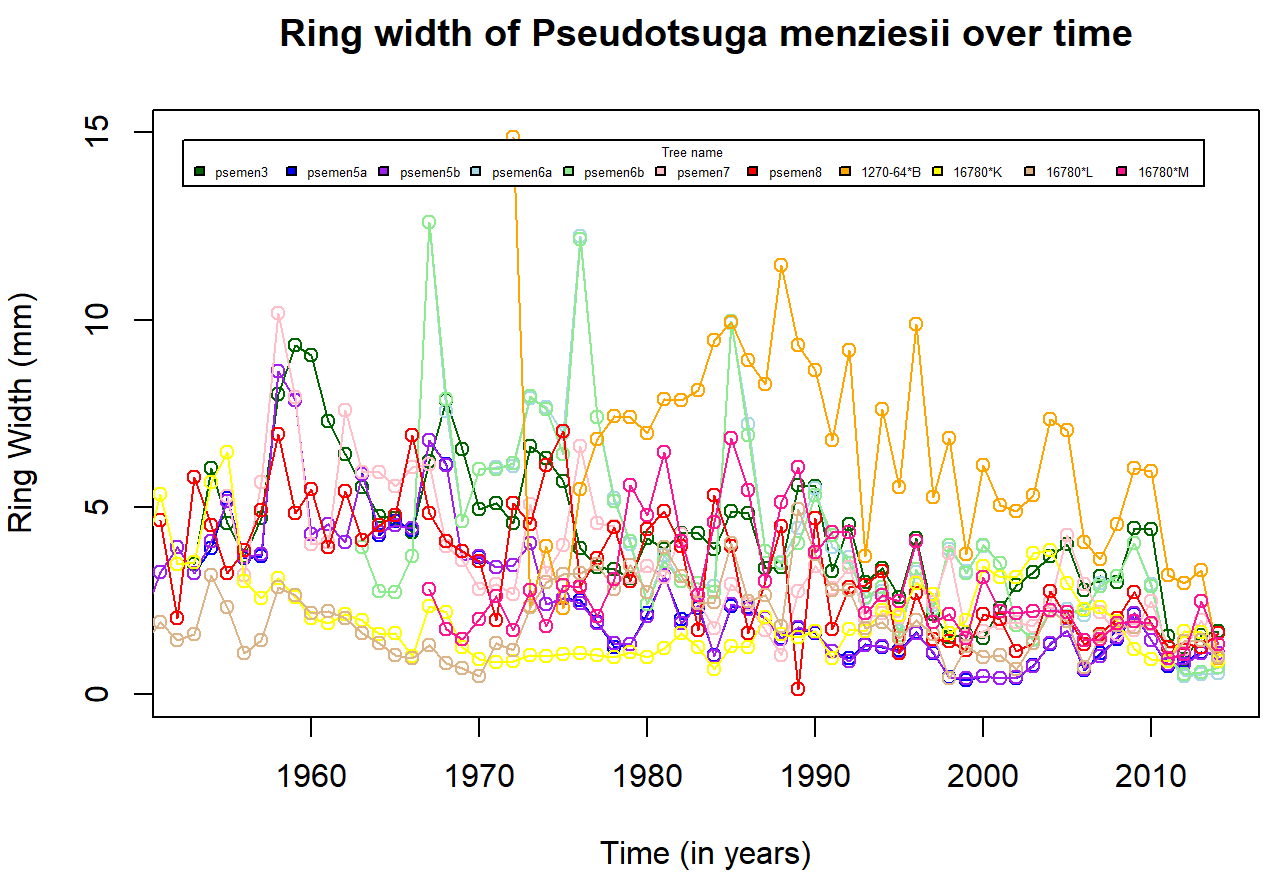
```
lines(thun5$Year, y = thun5$Ringwidth, type = "o", col = "lightgreen")

legend("topright", inset = .05, title = "Tree name", c("11371*G", "11371*J", "11371*M", "16592*D"
```



Pseudotsuga menziesii

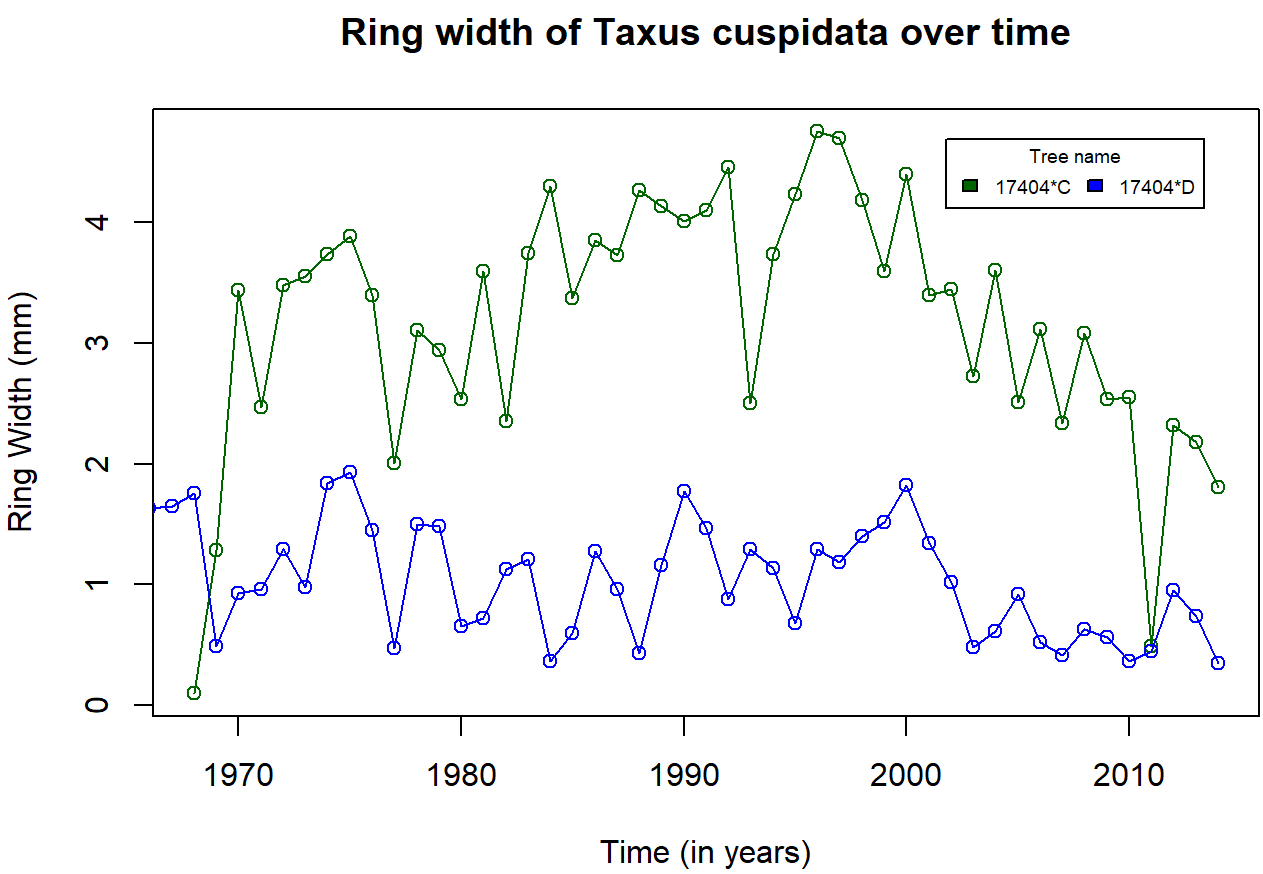
```
pseud1 <- subset(gymno_data_long, TreeName == "psemen3")

pseud2 <- subset(gymno_data_long, TreeName == "psemen5a")

pseud3 <- subset(gymno_data_long, TreeName == "psemen5b")

pseud4 <- subset(gymno_data_long, TreeName == "psemen6a")

pseud5 <- subset(gymno_data_long, TreeName == "psemen6b")

pseud6 <- subset(gymno_data_long, TreeName == "psemen7")

pseud7 <- subset(gymno_data_long, TreeName == "psemen8")

pseud8 <- subset(gymno_data_long, TreeName == "1270-64*B")
```

```r
pseud9 <- subset(gymno_data_long, TreeName == "16780*K")

pseud10 <- subset(gymno_data_long, TreeName == "16780*L")

pseud11 <- subset(gymno_data_long, TreeName == "16780*M")



plot(pseud1$Year, y = pseud1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", ylal
lines(pseud2$Year, y = pseud2$Ringwidth, type = "o", col = "blue")
lines(pseud3$Year, y = pseud3$Ringwidth, type = "o", col = "purple")
lines(pseud4$Year, y = pseud4$Ringwidth, type = "o", col = "lightblue")
lines(pseud5$Year, y = pseud5$Ringwidth, type = "o", col = "lightgreen")
lines(pseud6$Year, y = pseud6$Ringwidth, type = "o", col = "pink")
lines(pseud7$Year, y = pseud7$Ringwidth, type = "o", col = "red")
lines(pseud8$Year, y = pseud8$Ringwidth, type = "o", col = "orange")
lines(pseud9$Year, y = pseud9$Ringwidth, type = "o", col = "yellow")
lines(pseud10$Year, y = pseud10$Ringwidth, type = "o", col = "burlywood")
lines(pseud11$Year, y = pseud11$Ringwidth, type = "o", col = "deeppink")

legend("topright", inset = .05, title = "Tree name", c("psemen3", "psemen5a", "psemen5b", "psemen
```



Taxus cuspidata

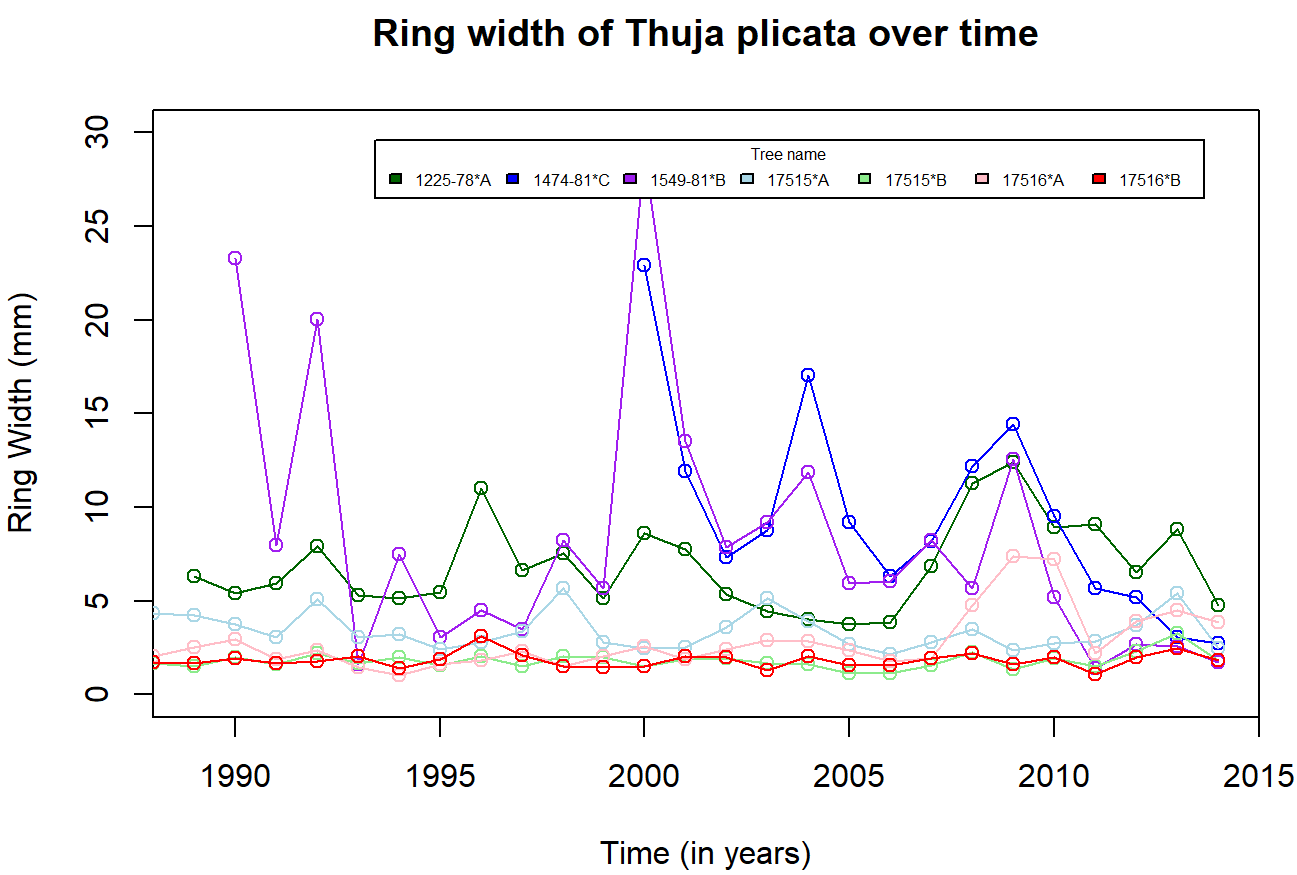
```
cusp1 <- subset(gymno_data_long, TreeName == "17404*C")

cusp2 <- subset(gymno_data_long, TreeName == "17404*D")


plot(cusp1$Year, y = cusp1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", ylab =
lines(cusp2$Year, y = cusp2$Ringwidth, type = "o", col = "blue")

legend("topright", inset = .05, title = "Tree name", c("17404*C", "17404*D"), fill = c("darkgreen
```



Thuja plicata

```
thuja1 <- subset(gymno_data_long, TreeName == "1225-78*A")

thuja2 <- subset(gymno_data_long, TreeName == "1474-81*C")

thuja3 <- subset(gymno_data_long, TreeName == "1549-81*B")

thuja4 <- subset(gymno_data_long, TreeName == "17515*A")
```

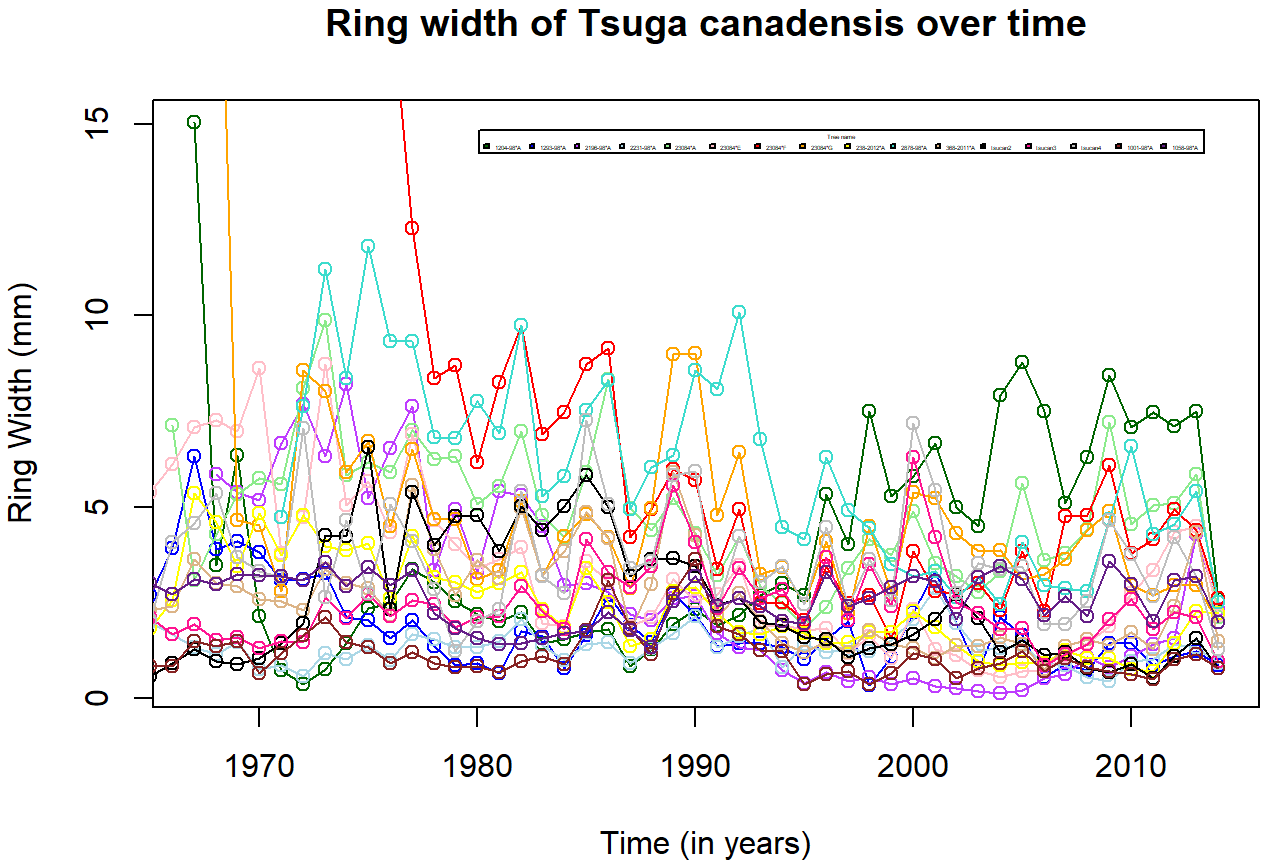
```
thuja5 <- subset(gymno_data_long, TreeName == "17515*B")

thuja6 <- subset(gymno_data_long, TreeName == "17516*A")

thuja7 <- subset(gymno_data_long, TreeName == "17516*B")


plot(thuja1$Year, y = thuja1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", ylal
lines(thuja2$Year, y = thuja2$Ringwidth, type = "o", col = "blue")
lines(thuja3$Year, y = thuja3$Ringwidth, type = "o", col = "purple")
lines(thuja4$Year, y = thuja4$Ringwidth, type = "o", col = "lightblue")
lines(thuja5$Year, y = thuja5$Ringwidth, type = "o", col = "lightgreen")
lines(thuja6$Year, y = thuja6$Ringwidth, type = "o", col = "pink")
lines(thuja7$Year, y = thuja7$Ringwidth, type = "o", col = "red")

legend("topright", inset = .05, title = "Tree name", c("1225-78*A", "1474-81*C", "1549-81*B", "17
```



Tsuga canadensis

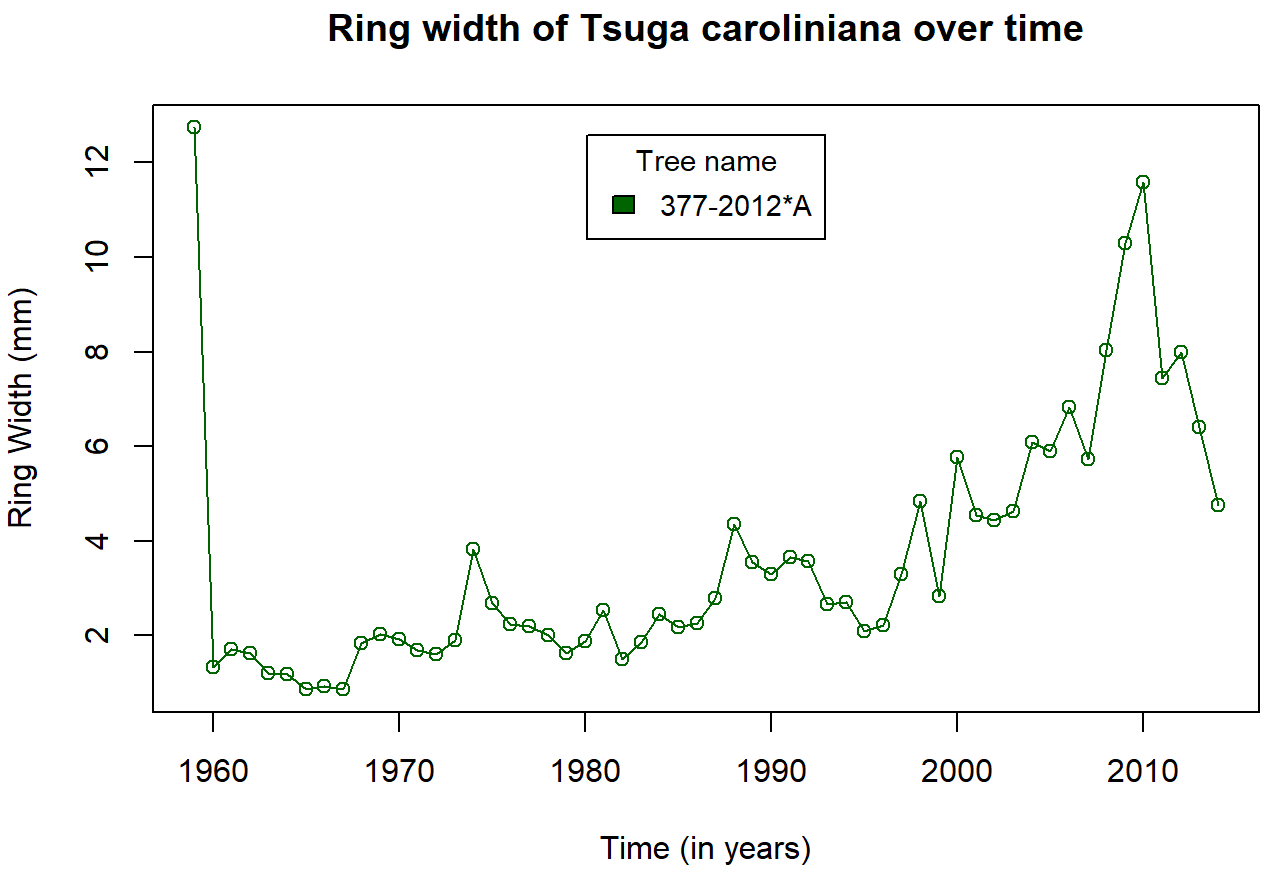
```
cana1 <- subset(gymno_data_long, TreeName == "1204-98*A")
```

```
cana2 <- subset(gymno_data_long, TreeName == "1293-98*A")

cana3 <- subset(gymno_data_long, TreeName == "2196-98*A")

cana4 <- subset(gymno_data_long, TreeName == "2231-98*A")

cana5 <- subset(gymno_data_long, TreeName == "23084*A")

cana6 <- subset(gymno_data_long, TreeName == "23084*E")

cana7 <- subset(gymno_data_long, TreeName == "23084*F")

cana8 <- subset(gymno_data_long, TreeName == "23084*G")

cana9 <- subset(gymno_data_long, TreeName == "238-2012*A")

cana10 <- subset(gymno_data_long, TreeName == "2878-98*A")

cana11 <- subset(gymno_data_long, TreeName == "368-2011*A")

cana12 <- subset(gymno_data_long, TreeName == "tsucan2")

cana13 <- subset(gymno_data_long, TreeName == "tsucan3")

cana14 <- subset(gymno_data_long, TreeName == "tsucan4")

cana15 <- subset(gymno_data_long, TreeName == "1001-98*A")

cana16 <- subset(gymno_data_long, TreeName == "1058-98*A")


plot(cana1$Year, y = cana1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", ylab =
lines(cana2$Year, y = cana2$Ringwidth, type = "o", col = "blue")
lines(cana3$Year, y = cana3$Ringwidth, type = "o", col = "darkorchid1")
lines(cana4$Year, y = cana4$Ringwidth, type = "o", col = "lightblue")
lines(cana5$Year, y = cana5$Ringwidth, type = "o", col = "lightgreen")
lines(cana6$Year, y = cana6$Ringwidth, type = "o", col = "pink")
lines(cana7$Year, y = cana7$Ringwidth, type = "o", col = "red")
lines(cana8$Year, y = cana8$Ringwidth, type = "o", col = "orange")
lines(cana9$Year, y = cana9$Ringwidth, type = "o", col = "yellow")
lines(cana10$Year, y = cana10$Ringwidth, type = "o", col = "turquoise")
lines(cana11$Year, y = cana11$Ringwidth, type = "o", col = "burlywood")
lines(cana12$Year, y = cana12$Ringwidth, type = "o", col = "black")
lines(cana13$Year, y = cana13$Ringwidth, type = "o", col = "deeppink")
lines(cana14$Year, y = cana14$Ringwidth, type = "o", col = "grey")
lines(cana15$Year, y = cana15$Ringwidth, type = "o", col = "brown4")
lines(cana16$Year, y = cana16$Ringwidth, type = "o", col = "darkorchid4")

legend("topright", inset = .05, title = "Tree name", c("1204-98*A", "1293-98*A", "2196-98*A", "22
```



Tsuga caroliniana

```
caroli1 <- subset(gymno_data_long, TreeName == "377-2012*A")


plot(caroli1$Year, y = caroli1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", y

legend("top", inset = .05, title = "Tree name", c("377-2012*A"), fill = c("darkgreen"), horiz = T
```


Ring width of Tsuga caroliniana over time
Tree name
377-2012*A
Ring Width (mm)
2
4
6
8
10
12
1960
1970
1980
1990
2000
2010
Time (in years)